Calibration along gradient descent in Gaussian tanh random-feature classifiers

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Abstract

We study ordinary full-batch gradient descent on cross-entropy in a two-layer Gaussian random-feature model with tanh activation and a logistic single-index teacher. At every fixed positive iteration and arbitrary positive finite sample/input and width/input ratios, an explicit causal recursion determines the asymptotic population risk and exact unbinned calibration conditional on the trained model. The proof verifies matrix universality for the actual nonlinear features and upgrades joint Gaussian approximation to total variation. In an explicit sample-rich regime, the calibration approximation is uniform over all positive times and the limiting estimator. In a separate strongly overparameterized regime, a large-norm witness gives a uniform calibration floor at every positive stopping time and a finite window for population-risk-optimal stopping. A scalar single-crossing argument proves strict average overconfidence at the risk-optimal first-step learning scale. All-time claims use stated sufficient ratio and step conditions; hidden features remain fixed throughout training.

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Introduction and relation to prior work

Calibration compares a predictor's stated probability with the label frequency conditional on that prediction. It is distinct from classification accuracy and from recovering the entire conditional probability given the input. Guo, Pleiss, Sun and Weinberger [13] documented the overconfidence of modern deep classifiers. This paper concerns exact score-conditional calibration for the realized trained network. The training matrix is reused at each update, and the algorithm is ordinary gradient descent.

Bai, Mei, Wang and Xiong [1] established high-dimensional overconfidence for logistic regression, including an underparameterized setting. Clarté, Loureiro, Krzakala and Zdeborová [2] already derived calibration and uncertainty descriptions for random-feature estimators. Thus neither random-feature calibration nor the Gaussian conditioning formula used below is a new subject or a new general formula. The contribution here is the discrete training trajectory of ordinary reused-data GD, exact model-conditional calibration convergence, and stopping-time conclusions under explicit assumptions.

Early-stopping benefits and calibration tradeoffs have also been studied in other scalings and models [6–8]. Our results keep proportional aspect ratios and a positive learning scale fixed. They do not rule out calibrated early stopping under different sample scalings, vanishing learning scales, feature learning, or post-training calibration.

The matrix spectral input is due to Louart, Liao and Couillet [3]; the rectangular memory and covariance formulas and their universality are due to Fan [4] and Wang, Zhong and Fan [5]. The appendix verifies their hypotheses for the present matrix and gives an exact auxiliary representation whose corrections cancel in the physical GD update. Theorem A covers every fixed finite horizon. Theorems B and D use different nonempty ratio regimes to control all positive stopping times. Theorem C concerns signed average overconfidence, which is stronger than positive calibration error and requires its stated choice of scale.

This is an internally reviewed research manuscript. Its precise contributions are presented without a claim of external acceptance or exclusive publication priority.

1. Model, algorithm, and calibration definition

Let $d, n, m \to \infty$, with

$$n/d \to \alpha \in (0, \infty), \qquad m/d \to \gamma \in (0, \infty).$$

Fix a deterministic unit vector θ_d, finite $\beta \geq 0$, and independent

$$X_i \sim N(0, I_d), \qquad W_j \sim N(0, I_d/d), \qquad Y_i \mid X_i \sim \text{Bernoulli}(\sigma(\beta \theta_d^\top X_i)),$$

where the labels are conditionally independent and $\sigma(u) = (1 + e^{-u})^{-1}$. The two-layer score and probability are

$$S_b(x) = \sum_{j=1}^{m} b_j \tanh(W_j^\top x), \qquad p_b(x) = \sigma(S_b(x)).$$

Only the output coefficients are trained. Write $H_{ij} = \tanh(W_j^\top X_i)$, and define mean cross-entropy

$$\mathcal{L}_n(b) = \frac{1}{n} \sum_i [\log(1 + e^{(Hb)_i}) - Y_i (Hb)_i].$$

For fixed $c > 0$, the **actual algorithm** is

$$b^0 = 0, \qquad b^{t+1} = b^t + \frac{cd}{nm} H^\top [Y - \sigma(Hb^t)]. \tag{1}$$

This is mean-loss GD with step cd/m in b, equivalently step cd in the normalized output coefficients $w = \sqrt{m}\, b$. The same sample and feature matrix are reused at every step.

Let $\mathcal{E}_d$ contain all training data and features. For a fresh independent input, put $Z = \theta_d^\top X$ and $S_{t,d} = S_{b^t}(X)$. Define

$$A_{t,d} = \mathbb{E}[Z S_{t,d} \mid \mathcal{E}_d], \qquad Q_{t,d} = \mathbb{E}[S_{t,d}^2 \mid \mathcal{E}_d].$$

The exact, unbinned, model-conditional calibration error is

$$\mathrm{CE}_{t,d} = \mathbb{E}\big[\, |\mathbb{E}[Y \mid S_{t,d}, \mathcal{E}_d] - \sigma(S_{t,d})| \mid \mathcal{E}_d \big]. \tag{2}$$

Conditioning on the score or its sigmoid is equivalent. This is not a fixed-bin estimator or an average of calibration computed under resampled models.

Throughout, $\|\mu - \nu\|_{\mathrm{TV}} = \sup_{|f|\le 1} |\int f\, d(\mu - \nu)|$, the full total-variation norm, twice the probability-distance convention.

For a centered Gaussian pair with $\mathrm{Var}\, Z = 1$, covariance A, and score variance $Q > A^2$, set

$$\begin{aligned} g_{A,Q}(s) &= \mathbb{E}_V \sigma\left(\beta\left[\frac{A}{Q} s + \sqrt{1 - A^2/Q}\, V\right]\right), \\ C_\beta(A,Q) &= \mathbb{E}_U \left| g_{A,Q}(\sqrt{Q}\, U) - \sigma(\sqrt{Q}\, U) \right|, \\ \mathcal{R}_\beta(A,Q) &= \mathbb{E}_U \log\left(2\cosh(\sqrt{Q}\, U/2)\right) - \mu_\beta A, \\ \mu_\beta &= \mathbb{E}[U(\sigma(\beta U) - 1/2)], \end{aligned} \tag{3}$$

where U, V are independent standard normals.

2. Finite-time theorem

Theorem A. Under the model above, for every fixed integer $T \ge 1$ and fixed $c > 0$, the explicit recursion in Section 3 defines deterministic A_t, Q_t and a positive-definite matrix Σ_T, with

$$Q_t - A_t^2 = (\Sigma_T)_{tt} > 0, \qquad 1 \le t \le T.$$

For each such time,

$$A_{t,d} \xrightarrow{P} A_t, \quad Q_{t,d} \xrightarrow{P} Q_t, \quad \left\| \mathcal{L}((Z, S_{t,d}) \mid \mathcal{E}_d) - N\left(0, \begin{pmatrix} 1 & A_t \\ A_t & Q_t \end{pmatrix}\right) \right\|_{\mathrm{TV}} \xrightarrow{P} 0. \tag{4}$$

The entire finite test-score path converges conditionally weakly in probability to the Gaussian path in (10) below. In particular,

$$\mathrm{CE}_{t,d} \xrightarrow{P} C_\beta(A_t, Q_t) > 0, \qquad \mathbb{E}[\ell(Y, S_{t,d}) \mid \mathcal{E}_d] \xrightarrow{P} \mathcal{R}_\beta(A_t, Q_t). \tag{5}$$

The TV and calibration convergences also hold in every finite L^p over the training environment, since their errors are uniformly bounded. The calibration curves satisfy the stronger integrated statement

$$\mathbb{E}\left[\, |\mathbb{E}[Y \mid S_{t,d}, \mathcal{E}_d] - g_{A_t,Q_t}(S_{t,d})| \mid \mathcal{E}_d \right] \xrightarrow{P} 0. \tag{6}$$

All finite aspect ratios are allowed, including $m > n$. The theorem fixes time before taking the dimension limit; it alone does not justify a growing-time conclusion.

3. Explicit recursion: the dependence on sample ratio, width, and time

Set

$$a = \mathbb{E}[U \tanh U], \quad v = \mathbb{E}[\tanh^2 U], \quad \nu = v - a^2 > 0, \quad \lambda = c/\alpha, \quad s = a\sqrt{\gamma}.$$

There are two scalar probability spaces. On the training-row space, $Z \sim N(0,1)$ and $Y \mid Z \sim \text{Bernoulli}(\sigma(\beta Z))$. Gaussian row fields $G_1, G_2, \ldots$ are independent of (Z, Y). On the column space, standard normal $H_\star$ is independent of Gaussian column fields. Both Gaussian covariance sequences are constructed below.

Initialize the training residual function $R_1 = Y - 1/2$, the column coefficient $q_0 = 0$, and empty matrices $\Psi_0, \Gamma_0, \Sigma_0$. At stage t, the previous row covariance determines the joint law of $(G_1, \ldots, G_{t-1})$ and hence the already defined functions $R_1, \ldots, R_t$. Compute

$$\widehat{\Delta}_t = \alpha(\mathbb{E}[R_i R_j])_{i,j \le t}, \quad (\widehat{\Phi}_t)_{ij} = \alpha \mathbb{E}[\partial_{G_j} R_i] \ (j < i), \quad \widehat{\ell}_t = \alpha \mathbb{E}[Z R_t], \tag{7}$$

with all other entries of $\widehat{\Phi}_t$ zero. Embed the preceding Ψ, Γ in $t \times t$ matrices with zero last row and column. First calculate

$$\begin{aligned} U_t &= (I - a^2 \widehat{\Phi}_t \Psi_t)^{-1}, \quad T_t = (I - a^2 \Psi_t \widehat{\Phi}_t)^{-1}, \\ \mathsf{B}_t &= \gamma^{-1}(\nu \widehat{\Phi}_t + a^2 U_t \widehat{\Phi}_t), \\ \Omega_t &= \gamma^{-1}\left[\nu \widehat{\Delta}_t + a^2 U_t (\widehat{\Delta}_t + a^2 \widehat{\Phi}_t \Gamma_t \widehat{\Phi}_t^\top) U_t^\top\right]. \end{aligned} \tag{8}$$

The last column of $\widehat{\Phi}_t$ is zero, so these expressions use only previous column information. If ψ_j denotes row j of Ψ, padded with zeros, complete the current row and coefficient by

$$\psi_t = \psi_{t-1} + \lambda \left[e_t^\top + \sum_{j<t} (\mathsf{B}_t)_{tj} \psi_j \right], \quad q_t = q_{t-1} + \lambda \left[\sum_{j<t} (\mathsf{B}_t)_{tj} q_j + \frac{s}{\gamma} \widehat{\ell}_t \right].$$

Here $\psi_0 = 0$. Put $q_{1:t} = (q_1, \ldots, q_t)^\top$ and

$$\begin{aligned} \Gamma_t &= \Psi_t \Omega_t \Psi_t^\top + q_{1:t} q_{1:t}^\top, \\ \mathsf{A}_t &= \nu \Psi_t + a^2 T_t \Psi_t, \\ \Sigma_t &= \nu \Gamma_t + a^2 T_t (\Gamma_t + a^2 \Psi_t \widehat{\Delta}_t \Psi_t^\top) T_t^\top, \end{aligned} \tag{9}$$

where T_t is now evaluated with the completed Ψ_t. Its inverses exist without a small-step assumption: the products with $\widehat{\Phi}_t$ are strictly lower triangular, so their inverses are finite geometric polynomials.

Let $G_{1:t} \sim N(0, \Sigma_t)$, consistently with previous stages and independently of (Z, Y), and set

$$S_t^{\text{train}} = G_t + \sum_{j \le t} (\mathsf{A}_t)_{tj} R_j + s q_t Z, \qquad R_{t+1} = Y - \sigma(S_t^{\text{train}}).$$

Derivatives required in (7) are computed recursively, with deterministic coefficients held fixed:

$$\partial_{G_j} R_{t+1} = -\sigma'(S_t^{\text{train}}) \left[\mathbf{1}_{j=t} + \sum_{k \le t} (\mathsf{A}_t)_{tk} \partial_{G_j} R_k \right].$$

This finishes one causal stage. Only finite matrices and Gaussian expectations occur. The fresh-test law and required overlaps are

$$S_t^{\text{test}} = s q_t Z + G_t, \quad A_t = s q_t, \quad Q_t = s^2 q_t^2 + (\Sigma_t)_{tt}. \tag{10}$$

Thus (3), (7)–(10) explicitly characterize calibration and risk as functions of $\alpha, \gamma, \beta, c, t$.

As an exact normalization check, at the first update

$$A_1 = ca^2\mu_\beta, \qquad Q_1 - A_1^2 = c^2\tau^2, \quad \tau^2 = \frac{\mu_\beta^2 a^2 v}{\gamma} + \frac{a^4 + v^2/\gamma}{4\alpha}. \tag{11}$$

This agrees with the direct one-step calculation in Appendix B. The two variance terms both matter.

4. Uniform training-time and early-stopping theorem

We give explicit sufficient constants, without claiming they are sharp. Let

$$L = 8\sqrt{1+\gamma}, \quad s_0 = \nu/2, \quad k_0 = \sqrt{s_0/2}, \quad p_0 = s_0^2/(20L^4),$$

$$B_0 = \sqrt{2\log(4/p_0)}, \quad p_1 = \sigma(-\beta B_0)p_0/2, \quad \mathfrak{a} = k_0 p_1/4,$$
$$C_* = \max\{1, (32L/(k_0 p_1))^2\}, \qquad \Lambda = 16L^2.$$

Assume additionally

$$\alpha/\gamma > C_*, \qquad 0 < c < \gamma/(2\Lambda). \tag{12}$$

These are finite constants for every fixed finite β and $\gamma > 0$, so the regime is nonempty. It has more samples than trained output coefficients.

Theorem B. Under (12), with probability tending to one, the empirical logistic loss has a unique finite MLE b^*, and GD converges to it geometrically with a dimension-independent contraction factor. Use $t = \infty$ for this MLE. The deterministic trajectory from Section 3 has limits A_∞, Q_∞, and

$$\sup_{t\in\{1,2,\ldots,\infty\}} \{|A_{t,d} - A_t| + |Q_{t,d} - Q_t|\} \xrightarrow{P} 0,$$

$$\sup_{t\in\{1,2,\ldots,\infty\}} |\mathrm{CE}_{t,d} - C_\beta(A_t, Q_t)| \xrightarrow{P} 0. \tag{13}$$

There is a deterministic $\varepsilon_* > 0$, depending only on the fixed parameters, such that

$$\mathbb{P}\left\{\inf_{t\in\{1,2,\ldots,\infty\}} \mathrm{CE}_{t,d} \geq \varepsilon_*/2\right\} \longrightarrow 1. \tag{14}$$

Consequently no training-measurable stopping rule with a positive integer value, or the value ∞, attains asymptotic calibration in this regime. A stopping rule may grow arbitrarily with dimension. If it diverges in probability, its calibration tends to $C_\beta(A_\infty, Q_\infty) > 0$. This conclusion keeps c fixed and positive; it does not cover dimension-dependent vanishing steps or all late-time overparameterized regimes.

At time zero, the constant prediction 1/2 is calibrated by teacher symmetry. Equation (14) concerns positive training times and nontrivial learning. It does not confuse calibration with Bayes accuracy.

5. Strict overconfidence with an improved predictor

Theorem C. Let $\beta > 0$ and any fixed $\alpha, \gamma > 0$. Put $k = a^2\mu_\beta > 0$, let $\tau > 0$ be given by (11), and set $U = kZ + \tau G$ with independent standard normals. Define $g(u) = \mathbb{E}[\sigma(\beta Z) \mid U = u]$ and

$$R(c) = \mathbb{E}[\log(1 + e^{cU}) - cUg(U)].$$

There is a unique population-risk-optimal first-step scale $c_* > 0$. For GD (1) run with this fixed scale, its first-step population risk and signed overconfidence satisfy

$$\mathcal{R}_{1,d} \xrightarrow{P} R(c_*) < \log 2,$$

$$\mathrm{OC}_{1,d} := \mathbb{E}[(\sigma(S_{1,d}) - \sigma(\beta Z))\,\mathrm{sign}(S_{1,d}) \mid \mathcal{E}_d] \xrightarrow{P} \mathrm{OC}_* > 0. \tag{15}$$

The signed quantity is **average predicted confidence minus classification accuracy**. It does not assert pointwise overconfidence at every score. In fact the optimal-scale Gaussian predictor is overconfident close to its decision boundary and underconfident in the far tails.

This theorem uses Theorem A with $c = c_*$. The scale need not automatically lie in (12); the following corollary gives explicit sufficient conditions for that compatibility. Optimizing first-step risk is not asserted to optimize over the entire GD path.

Corollary: one fixed model can satisfy all three conclusions. The regimes are not disjoint. Retain L, k_0, p_0, B_0 above, and choose

$$\alpha/\gamma > C_0 := \max\{1, (256L/(k_0p_0))^2\}, \qquad 0 < \beta < \min\{\log(3)/B_0,\ v/(8aL\sqrt{\alpha})\}. \tag{16}$$

Then $\sigma(-\beta B_0) > 1/4$, so $C_*(\beta) \le C_0$. Further,

$$c_* < \frac{\beta k}{k^2 + \tau^2} \le \frac{a^2\alpha\gamma\beta^2}{v^2} < \frac{\gamma}{64L^2} < \frac{\gamma}{2\Lambda}.$$

Here $\mu_\beta \le \beta/4$ and $\tau^2 \ge v^2/(4\alpha\gamma)$. Thus the **same** fixed-step network, with $c = c_*$, has the complete deterministic training-time calibration curve, an improved and strictly overconfident first iterate, and the uniform positive calibration floor for every positive stopping time. Equation (16) is an explicit nonempty open regime with a weak but nonzero teacher. It does not impose this restriction on Theorem A or on Theorem C separately.

6. All positive stopping times in a strongly overparameterized regime

There is a second all-time regime, with more trained coefficients than samples. Its proof does not require a late-time Gaussian approximation. Define, with ν as in Section 3,

$$s_X = \nu/2, \quad \kappa_X = \sqrt{s_X/2}, \quad L_X = 8\sqrt{1+\alpha}, \quad p_X = s_X^2/(20L_X^4),$$

$$C_X = \max\{1, (32L_X/(\kappa_X p_X))^2\}, \qquad k_X = \kappa_X^2 p_X/16, \quad K_X = 64L_X^2.$$

Assume

$$\gamma/\alpha > C_X, \qquad 0 < c < 2\alpha/K_X. \tag{17}$$

For every fixed $\alpha > 0$ and finite β, this is a nonempty proportional regime. The constants are conservative sufficient conditions.

Theorem D. Under (17), there is $\varepsilon > 0$ such that

$$\mathbb{P}\{\inf_{t\in\{1,2,\dots\}} \mathrm{CE}_{t,d} \geq \varepsilon\} \longrightarrow 1. \tag{18}$$

Thus every training-measurable stopping rule taking a finite positive integer value, including one diverging arbitrarily with dimension, has a positive exact-calibration floor. The empirical loss converges to zero, the coefficient norms diverge, and population cross-entropy diverges uniformly at late times: for every $M < \infty$, there is a deterministic T_M with

$$\mathbb{P}\{\inf_{t\geq T_M} \mathcal{R}_{t,d} > M\} \longrightarrow 1. \tag{19}$$

There is a deterministic finite T_0 such that, with probability tending to one, every population-risk minimizing time on the entire GD path lies in $\{0, \dots, T_0\}$. Theorem A therefore determines the limiting optimal risk by minimizing over this finite window. If the limiting minimizer is unique, the minimizing time converges to it. For $\beta > 0$, a sufficiently small fixed positive c gives a first-step improvement over the constant predictor while retaining the calibration obstruction.

Proof. Appendix F proves that $HH^\top/m$ lies between $k_X I$ and $K_X I$ with probability tending to one. A separating comparator and the convex GD inequality give a dimension-uniform loss upper bound tending to zero. Jensen's inequality then forces every late iterate's norm above any specified constant. A uniform small-ball event for fresh features and finite teacher noise gives a score-measurable witness to calibration error for every sufficiently large coefficient vector, whether delocalized or not. Combining that witness with Theorem A for the finitely many earlier times proves (18). A related witness lower bounds population cross-entropy by a positive multiple of coefficient norm, proving (19) and the finite optimization window. The full inequalities, constants, and quantifiers are in Appendix F.

Time ∞ is not a finite MLE in this separable regime, and no claim is made here about the calibration of a terminal hard classifier. Theorem D supplies a uniform lower bound, not an exact late-time calibration curve. The exact curve in Theorem A fixes time before dimension. Theorem B gives a uniform curve under its separate assumptions.

7. Proof of the combined theorem

The detailed proofs are in Appendices A–F. The following proof establishes their logical composition and identifies the imported mathematical results.

Step 1: actual random-feature matrix and independent teacher signal. Rotate $\theta = e_1$, and write input and feature vectors as their first coordinate plus perpendicular coordinates. Taylor expansion of tanh gives, for training rows and a proportional independent test block together,

$$H/\sqrt{m} = A_0 + (a\sqrt{\gamma}/m)zh^\top + E_d, \qquad \|E_d\|_{\mathrm{op}} \xrightarrow{P} 0.$$

The perpendicular matrix A_0 is independent of z, h, labels, and the training/test mask. The error follows from Gaussian sup-norm bounds, a centered derivative-feature matrix bound, and a Frobenius bound on the Taylor remainder; no entrywise-to-operator inference is omitted.

Step 2: the universality hypotheses are proved for this matrix. Clip each underlying Gaussian matrix in operator norm and then clip the normalized feature matrix. These Frobenius-nonexpansive projections preserve signed-permutation symmetry and make each fixed matrix-power entry globally $O(d^{-1/2})$-Lipschitz in all Gaussian entries. Gaussian concentration gives summable failure bounds for every fixed power: off-diagonal entries vanish at the

required rate, and diagonal entries approach their normalized trace uniformly. The clipped and original matrices agree eventually almost surely. Conditional Hermite covariance analysis and the Louart–Liao–Couillet spectral theorem identify the limiting singular law with the biorthogonally invariant Gaussian comparison matrix. These facts verify Wang–Zhong–Fan Proposition D.1(b2), including more than just an empirical spectral distribution.

The row-normalized rectangular cumulants are

$$\kappa_2 = v, \qquad \kappa_{2j} = a^{2j}\delta^{j-1} \quad (j \geq 2),$$

where δ is the total training-plus-query row ratio before the query block is eliminated. Substitution into Fan's rectangular covariance and memory formulas sums finite triangular series. After conditioning on training rows, every δ cancels exactly, yielding (8)–(9).

Step 3: ordinary GD is represented causally. Write $w^t = \sqrt{m}\, b^t$ for the normalized coefficient vector. Introduce auxiliary column fields by subtracting the deterministic memory terms B from $A_0^\top \mathbf{R}_t$, and row fields by subtracting A from $A_0 w^t$, where $\mathbf{R}_t$ is the sample residual vector. Define the coordinatewise nonlinearities by adding exactly these terms back. They cancel algebraically in the physical update. Teacher overlaps are first fixed to their deterministic recursive expectations. The resulting history-dependent maps meet the continuity, growth, Lipschitz, and independent-side-information assumptions of Wang–Zhong–Fan Theorem 2.22.

No degeneracy assumption is left open. The residual second-moment matrix begins with $\Delta_1 > 0$. From (8)–(9), Ω_t dominates a positive multiple of Δ_t, and $\Sigma_t \succeq \nu\Gamma_t$. The derivative matrix Ψ_t has positive diagonal λ, so $\Gamma_t \succ 0$. Conditional on the past and a training row, the next sigmoid residual depends strictly on a fresh Gaussian innovation; hence it cannot lie in the linear span of preceding residuals. This proves positive definiteness inductively for every fixed time and every $c > 0$.

The AMP theorem now gives joint empirical state convergence. Its overlap convergence, a normalized Euclidean stability bound, and a finite induction replace frozen teacher overlaps by the actual empirical ones. The same stability argument removes E_d. On query rows all residuals are zero, so the limiting physical score is exactly (10). Conditional laws of large numbers and conditional fourth-moment bounds identify the realized-network covariances. This proves finite-time conditional weak convergence and the deterministic overlaps for the original nonlinear GD, rather than a different AMP training algorithm.

Step 4: exact calibration requires, and receives, total variation. A leave-one-feature comparison proves $\|b^t\|_2 = O_P(1)$ and $\|b^t\|_\infty = O_P(\sqrt{\log d/d})$ at any fixed horizon. Thus the fresh score has bounded gradient and Hessian Hilbert–Schmidt norm, while its Hessian operator norm vanishes. An Ornstein–Uhlenbeck covariance identity proves Gaussian weak approximation with matched covariance. The teacher-orthogonal derivative Gram determinant is concentrated away from zero whenever the residual score variance is positive. Localized inverse-vector fields and Gaussian integration by parts give a dimension-free total-variation smoothing bound. Mollification then upgrades weak convergence to joint TV. This analytic argument is proved in Appendix C.

For any joint law μ of (Z, S), define the signed measure

$$\zeta_\mu(B) = \int \mathbf{1}_{s\in B}[\sigma(\beta z) - \sigma(s)]\, \mu(dz, ds).$$

Its TV norm equals calibration error. Therefore $|C(\mu) - C(\widetilde{\mu})| \leq \|\mu - \widetilde{\mu}\|_{\mathrm{TV}}$. The same measure argument controls the integrated conditional regression curve. Logistic loss is at most $\log 2 + |S|$, so bounded conditional second moments justify truncation for risk. These observations prove (4)–(6).

Strict positivity in (5) is elementary. For $A \le 0$ or $\beta = 0$, the conditional teacher curve cannot equal a strictly increasing nonconstant sigmoid. For $A > 0, \beta > 0$, write it as $g(s) = \mathbb{E}\sigma(ks+rG)$ with $r > 0$. Its upper tail satisfies $e^{ks}(1 - g(s)) \to e^{r^2/2}$. Equality with $\sigma(s)$ would first require $k = 1$, then $r = 0$, a contradiction.

Step 5: all times are controlled on one event. Under (12), the conditional feature covariance and its teacher-orthogonal residual both dominate $(\nu/2)I$. A bounded-cutoff small-ball argument and the noisy labels yield empirical coercivity uniformly in every coefficient direction. Uniform unsigned small-ball bounds give a sample covariance lower bound even after deleting an arbitrary small fraction of rows. On the bounded initial loss sublevel set, this yields uniform positive logistic Hessian curvature. GD and every feature-deleted GD path stay in that same ball and contract by a common dimension-independent factor $q < 1$.

Feature-deleted minimizers are defined on a fixed compact ball even on exceptional data, preserving independence from the omitted feature. Their missing-coordinate gradients have centered Gaussian concentration. A union bound only through a deterministic $O(\log d)$ horizon, followed by the common geometric tail, proves

$$\sup_{t \ge 0} \|b^t\|_\infty = O_P(\sqrt{\log d / d}).$$

The first update makes a nonvanishing loss decrease, so convexity bounds every positive iterate's norm away from zero. This gives uniform lower residual variance and upper score variance. The TV theorem is sequentially uniform on this bounded class. Its calibration functional is continuous and strictly positive on the resulting compact covariance set, proving (14).

Finally, the geometric coefficient tail gives a geometric tail for $A_{t,d}, Q_{t,d}$. Fixed-time deterministic limits from Theorem A are therefore exponentially Cauchy. Splitting a supremum into finitely many times and the common tail proves (13). This supplies the needed exchange of training-time and dimension limits rather than assuming it.

Step 6: signed overconfidence at the best first-step scale. For $U = kZ + \tau G$, Gaussian conditioning gives $g(u) = \mathbb{E}\sigma(\lambda_0 u + rG)$, where $\lambda_0 = \beta k/(k^2 + \tau^2) > 0$ and $r > 0$. Appendix E proves that $\operatorname{logit} g(u)/u$ is strictly increasing on $u > 0$, from $4g'(0)$ to λ_0. Its proof derives a normal variance-mixture representation of logistic noise from its characteristic function, then proves the necessary strict single-crossing property by an explicit kernel-ratio inequality.

The cross-entropy has strictly positive second derivative and a unique minimizer $c_* \in (4g'(0), \lambda_0)$. The error $e(u) = \sigma(c_* u) - g(u)$ is positive below one threshold u_* and negative above it. Risk stationarity gives $\mathbb{E}[Ue(U)] = 0$. Consequently

$$\mathrm{OC}_* = 2 \int_0^\infty e(u)(1 - u/u_*)\varphi_{k^2+\tau^2}(u)\, du > 0.$$

Both factors have the same strict sign. Also $R'(0) < 0$, so $R(c_*) < \log 2$. Theorem A's TV and risk convergence transfer both strict conclusions to the actual trained network, proving (15). This proves Theorems A–C. Appendix F proves Theorem D by a separate large-norm argument, as described in Section 6.

Appendix A. Feature matrices and the finite-time GD representation

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

A.1 Model and normalization

Use the Gaussian model of Section 1: inputs $X_i \sim N(0, I_d)$, a deterministic unit teacher θ, conditionally independent labels $Y_i \mid X_i \sim \text{Bernoulli}(\sigma(\beta\theta^\top X_i))$, fixed finite $\beta \geq 0$, and independent feature rows $W_j \sim N(0, I_d/d)$. Let $n/d \to \alpha > 0, m/d \to \gamma > 0$, $\phi = \tanh$, $H_{ij} = \phi(W_j^\top X_i)$, and $K = HH^\top/m$.

Write the score as $S_t(x) = (b^t)^\top h(x)$, where $b^t = a^t/\sqrt{m}$ in the old network convention. Mean-loss GD with output-weight step cd is exactly

$$b^0 = 0, \quad u^t = Hb^t, \quad r^t = Y - \sigma(u^t), \quad b^{t+1} = b^t + \frac{\lambda_d}{m} H^\top r^t, \qquad \lambda_d = \frac{cd}{n} \longrightarrow \frac{c}{\alpha}. \tag{A.1.1}$$

Consequently $u^{t+1} = u^t + \lambda_d K r^t$. All statements below allow any fixed $c > 0$, except the explicit small-step conclusion in Proposition 3. The horizon T is a fixed integer independent of dimension. No claim concerns $T_d \to \infty$.

Define

$$a = \mathbb{E}[G\phi(G)], \quad v = \mathbb{E}\phi(G)^2, \quad \nu = v - a^2 > 0, \quad A(q) = \mathbb{E}\phi'(\sqrt{q}G). \tag{A.1.2}$$

Strict positivity of ν follows because tanh is not a scalar multiple of its input in Gaussian L^2.

A.2 Uniform feature-matrix bounds

Conditional on F, whose rows are $W_j^\top$, the rows of H are independent centered random vectors. For any unit $v_0 \in \mathbb{R}^m$, the function $x \mapsto v_0^\top \phi(Fx)$ is $\|F\|_{\text{op}}$-Lipschitz and odd. Gaussian concentration therefore gives

$$\mathbb{E}[\exp(sv_0^\top h(X)) \mid F] \leq \exp(s^2\|F\|_{\text{op}}^2/2). \tag{A.2.1}$$

For a unit $u_0 \in \mathbb{R}^n$, conditional independence gives the same bound for $u_0^\top H v_0$. Taking 1/4-nets of the two unit spheres and using the standard net inequality proves

$$\|H\|_{\text{op}} = O_P(\sqrt{d}), \qquad \|K\|_{\text{op}} = O_P(1). \tag{A.2.2}$$

The failure probabilities at sufficiently large fixed constants decay exponentially in d. The same net argument for the Gaussian matrices proves $\|F\|_{\text{op}} = O_P(1)$ and $\|X\|_{\text{op}} = O_P(\sqrt{d})$, again with exponential tails. Integrating these bounds shows that all fixed moments of the corresponding normalized norms are uniformly bounded.

The elementary proof above does not infer a bounded spectral norm merely from entrywise boundedness.

A.3 Delocalization along every fixed finite GD path

Theorem 1. For each fixed $T < \infty$,

$$\max_{0\le t\le T} \|b^t\|_2 = O_P(1), \qquad \max_{0\le t\le T} \|b^t\|_\infty = O_P\left(\sqrt{\frac{\log d}{d}}\right). \tag{A.3.1}$$

The first bound has uniformly bounded fixed moments. The proof uses the same reused training sample at every step.

Proof. Since $\|r^t\|_2 \le \sqrt{n}$, (A.1.1) gives directly

$$\|b^t\|_2 \le \frac{T\lambda_d}{m}\|H\|_{\mathrm{op}}\sqrt{n} = O_P(1). \tag{A.3.2}$$

For the coordinate bound, denote feature column j by h_j, replace that column by zero, and retain the denominator m. Write $K^{-j} = K - h_j h_j^\top/m$, and let $u^{-j,t}, r^{-j,t}$ be the corresponding GD scores and residuals. The matrices satisfy $0 \preceq K^{-j} \preceq K$.

Conditional on X, Y, F_{-j}, the vector $r^{-j,t}$ is independent of W_j, bounded in Euclidean norm by $\sqrt{n}$, and

$$G \mapsto (r^{-j,t})^\top \phi(XG/\sqrt{d})$$

is odd and $\|X\|_{\mathrm{op}}\sqrt{n/d}$-Lipschitz for $G \sim N(0, I_d)$. Its conditional mean is zero. Gaussian concentration followed by a union bound over $j \le m, t < T$, on $\|X\|_{\mathrm{op}} \le C\sqrt{d}$, therefore yields

$$\max_{j,t<T} |h_j^\top r^{-j,t}| = O_P(\sqrt{d\log d}). \tag{A.3.3}$$

The conditioning here is essential: later residuals in the original model are not independent of W_j.

Set $e^{j,t} = u^t - u^{-j,t}$. Since sigmoid is 1/4-Lipschitz,

$$\|e^{j,t+1}\|_2 \le \left(1 + \frac{\lambda_d\|K\|_{\mathrm{op}}}{4}\right)\|e^{j,t}\|_2 + \frac{\lambda_d\|h_j\|_2}{m}|h_j^\top r^{-j,t}|. \tag{A.3.4}$$

We have $e^{j,0} = 0$ and $\|h_j\|_2 \le \sqrt{n}$. On the simultaneous high-probability bounds (A.2.2) and (A.3.3), iteration for fixed T gives

$$\max_{j,t\le T} \|e^{j,t}\|_2 = O_P(\sqrt{\log d}). \tag{A.3.5}$$

Thus

$$|h_j^\top r^t| \le |h_j^\top r^{-j,t}| + \tfrac{1}{4}\sqrt{n}\,\|e^{j,t}\|_2 = O_P(\sqrt{d\log d})$$

uniformly over the finitely many times and all features. Summing (A.1.1) over time and dividing by m proves the second assertion in (A.3.1). □

A.4 Joint fresh-test Gaussian approximation for the entire finite path

Let $\mathcal{E}_d$ contain the features and training sample, and define the exact conditional covariances

$$A_{t,d} = \mathbb{E}_X[(\theta^\top X)S_t(X) \mid \mathcal{E}_d], \qquad Q_{st,d} = \mathbb{E}_X[S_s(X)S_t(X) \mid \mathcal{E}_d]. \tag{A.4.1}$$

Write C_d for the covariance matrix of $(\theta^\top X, S_1(X), \dots, S_T(X))$. All means are zero by oddness.

Theorem 2. For fixed T, in bounded-Lipschitz distance,

$$d_{\mathrm{BL}}\Big(\mathcal{L}((\theta^\top X, S_1(X), \dots, S_T(X)) \mid \mathcal{E}_d), N(0, C_d)\Big) \longrightarrow 0 \quad \text{in probability.} \tag{A.4.2}$$

This is a Gaussian approximation whose covariance is allowed to depend on the realized network. It does not assert that C_d has a deterministic limit.

Proof. Fix real $u, v_1, \dots, v_T$, put $b = \sum_t v_t b^t$, and let

$$f(x) = u\theta^\top x + b^\top \phi(Fx).$$

Then $\mathbb{E}f(X) = 0$, and globally

$$\|\nabla f\|_2 \le L := |u| + \|F\|_{\mathrm{op}}\|b\|_2 = O_P(1), \quad \|\nabla^2 f\|_{\mathrm{op}} \le \varepsilon := \|\phi''\|_\infty \|F\|_{\mathrm{op}}^2 \|b\|_\infty = o_P(1). \tag{A.4.3}$$

Here is a self-contained normal-approximation argument. For the Gaussian Ornstein–Uhlenbeck semigroup P_s, set

$$U_f(x) = \int_0^\infty e^{-s} P_s \nabla f(x)\, ds, \qquad T_f(x) = \nabla f(x) \cdot U_f(x).$$

Gaussian integration by parts, or integration of $\frac{d}{ds}P_s f$, gives for smooth bounded g

$$\mathbb{E}[f(X)g(f(X))] = \mathbb{E}[T_f(X)g'(f(X))], \quad \mathbb{E}T_f = \mathbb{E}f^2 =: q. \tag{A.4.4}$$

The semigroup derivative identity gives $\|U_f\|_\infty \le L$ and $\|DU_f\|_{\mathrm{op},\infty} \le \varepsilon/2$, whence $\|\nabla T_f\|_2 \le 3L\varepsilon/2$. Gaussian Poincaré yields

$$\mathbb{E}|T_f - q| \le \tfrac{3}{2} L\varepsilon. \tag{A.4.5}$$

For $\psi(t) = \mathbb{E}e^{itf(X)}$, (A.4.4) gives $\psi'(t) = -tq\psi(t) - t\mathbb{E}[(T_f - q)e^{itf(X)}]$. Solving this scalar differential equation and using $q \ge 0$ gives

$$|\psi(t) - e^{-qt^2/2}| \le \tfrac{3}{4} t^2 L\varepsilon. \tag{A.4.6}$$

No lower bound on q is needed.

The conditional covariance matrices are tight because (A.4.3) and Poincaré bound all diagonal entries. Every subsequence admits a further subsequence on which the bounds in (A.4.3) hold almost surely and C_d converges along further deterministic realizations in a compact set. Applying (A.4.6) to a countable dense collection of coefficient vectors, tightness and the characteristic-function criterion identifies every conditional subsequential limit with the corresponding Gaussian covariance limit. This proves (A.4.2). □

The old covariance-operator lemma applies uniformly to these data-dependent weights because their Euclidean norms are tight:

$$A_{t,d} = a\theta^\top F^\top b^t + o_P(1), \quad Q_{st,d} = a^2 (F^\top b^s)^\top (F^\top b^t) + \nu (b^s)^\top b^t + o_P(1). \tag{A.4.7}$$

Indeed the exact first formula is $A_{t,d} = \theta^\top F^\top D_A b^t$, where $D_A = \mathrm{diag}(A(\|W_j\|^2))$, and $\|D_A - aI\|_{\mathrm{op}} \to 0$. The exact second formula is $Q_{st,d} = (b^s)^\top \Sigma_F b^t$.

For any subsequence on which $(A_{t,d}, Q_{st,d})$ converges in probability to deterministic limits, Theorem 2 supplies the full joint finite-time Gaussian test-law limit on that subsequence. With fresh labels, apply the common Markov kernel $Y \mid Z \sim \mathrm{Bernoulli}(\sigma(\beta Z))$.

The total-variation and exact-calibration strengthening is proved separately in Appendix C; its conditions are furnished by Theorem 1 together with a nonvanishing coefficient norm, for example Proposition 3 below.

A.5 A rigorous nondegeneracy condition

Put $\rho_\beta = \mathbb{E}\sigma'(\beta G) > 0$. This is positive for every fixed finite β.

Proposition 3.

$$\|b^1\|_2^2 \geq \frac{c^2 v \rho_\beta}{\alpha\gamma} + o_P(1). \tag{A.5.1}$$

If $\lambda_d \|K\|_{\mathrm{op}} \leq 4$, then pathwise

$$\|b^t\|_2 \geq \tfrac{1}{2}\|b^1\|_2 \quad \text{for every integer } t \geq 1. \tag{A.5.2}$$

In particular a sufficiently small, fixed positive c makes (A.5.2) hold with probability tending to one. At every fixed positive time in this regime, the teacher-orthogonal fresh-score variance is bounded away from zero in probability.

Proof. Conditional on X, F, write $R = Y - 1/2 = \mu + \epsilon$, where $\mu_i = \sigma(\beta Z_i) - 1/2$, independent centered ϵ_i have variance $D_{ii} = \sigma'(\beta Z_i)$, and D is diagonal. Since

$$\|b^1\|_2^2 = \frac{\lambda_d^2}{m} R^\top K R,$$

expanding the quadratic form and using independence and bounded fourth moments gives

$$\mathrm{Var}(R^\top K R \mid X, F) \leq C\{\mathrm{tr}(K^2) + \|K\mu\|_2^2\} \leq Cn\|K\|_{\mathrm{op}}^2. \tag{A.5.3}$$

Thus $\|b^1\|_2^2 = \lambda_d^2 m^{-1}(\mu^\top K\mu + \mathrm{tr}\, KD) + o_P(1)$. The first term is nonnegative. Conditional on F, the row quantities $D_{ii}K_{ii}$ are iid, bounded, and have mean

$$\frac{1}{m}\sum_j \mathbb{E}_X[\sigma'(\beta\theta^\top X)\phi(W_j^\top X)^2].$$

Uniformly over j, $\|W_j\|^2 \to 1$ and $\theta^\top W_j \to 0$. Continuity of bounded Gaussian expectations therefore makes this mean tend to $v\rho_\beta$. The conditional LLN proves $n^{-1}\,\mathrm{tr}\, KD \to v\rho_\beta$, yielding (A.5.1).

For (A.5.2), let $L(b) = \sum_i \ell(Y_i, h_i^\top b)$, $g_0 = \nabla L(0) = -H^\top R$, and $\eta = \lambda_d/m$. Its gradient is $\|H\|_{\mathrm{op}}^2/4$-Lipschitz. Under the displayed step condition, descent gives $L(b^t) \leq L(b^1) \leq L(0) - \eta\|g_0\|^2/2$. Convexity gives $L(b^t) \geq L(0) + g_0^\top b^t$. Combining these and Cauchy–Schwarz yields $\|b^t\| \geq \eta\|g_0\|/2 = \|b^1\|/2$. Finally, the exact Hermite decomposition implies

$$Q_{tt,d} - A_{t,d}^2 \geq (\nu - o_P(1))\|b^t\|_2^2, \tag{A.5.4}$$

uniformly for the fixed-time path. □

A.6 Teacher-coordinate removal is operator-small

Rotate coordinates so that $\theta = e_1$. Write $X_i = (z_i, x_i)$ and $W_j = (t_j, f_j)$, with $x_i \in \mathbb{R}^{d-1}$, $f_j \in \mathbb{R}^{d-1}$. Let V have rows $f_j^\top$, $U_{ij} = x_i^\top f_j$, $H^0 = \phi(U)$. The vectors z, t, the noise matrix H^0, and label randomness conditional on z have the required independence: H^0 is independent of (z, t, Y).

Proposition 4. For any fixed proportional number of rows, including training rows together with an additional proportional block of test rows,

$$\left\| \frac{H - H^0 - azt^\top}{\sqrt{m}} \right\|_{\mathrm{op}} \longrightarrow 0 \quad \text{in probability.} \tag{A.6.1}$$

Proof. Taylor's theorem gives

$$H - H^0 - azt^\top = \operatorname{diag}(z)[\phi'(U) - a\mathbf{1}\mathbf{1}^\top]\operatorname{diag}(t) + R, \quad |R_{ij}| \le Cz_i^2 t_j^2. \tag{A.6.2}$$

The remainder satisfies

$$\|R\|_F^2 \le C\Big(\sum_i z_i^4\Big)\Big(\sum_j t_j^4\Big) = O_P(1),$$

so $\|R\|_{\mathrm{op}}/\sqrt{m} = o_P(1)$.

Conditional on V, the centered rows of the derivative matrix $\phi'(U)$ are independent. The same concentration-and-net proof as in Section A.2 applies with Lipschitz constant $\|\phi''\|_\infty \|V\|_{\mathrm{op}}$; hence the centered matrix has operator norm $O_P(\sqrt{d})$. Its row mean minus $a\mathbf{1}$ is the vector $\delta_j = A(\|f_j\|^2) - a$. Bounded A' and $\mathbb{E}(\|f_j\|^2 - 1)^2 = O(d^{-1})$ give $\|\delta\|_2 = O_P(1)$. Thus

$$\|\phi'(U) - a\mathbf{1}\mathbf{1}^\top\|_{\mathrm{op}} = O_P(\sqrt{d}).$$

Finally $\|z\|_\infty = O_P(\sqrt{\log d})$ and $\|t\|_\infty = O_P(\sqrt{\log d/d})$, so the first matrix in (A.6.2), after division by $\sqrt{m}$, is $O_P(\log d/\sqrt{d}) = o_P(1)$. □

This is an operator-norm statement for the normalized *feature matrix*, not just for its population covariance. If $A, \widetilde{A}$ have bounded operator norm and $\|A - \widetilde{A}\|_{\mathrm{op}} = o_P(1)$, finite-time logistic GD in the parameter $w = \sqrt{m}b$, with normalized scores Aw, differs by $o_P(\sqrt{m})$ in coefficient norm and $o_P(\sqrt{n})$ in score norm. To verify, subtract the two recursions, use 1/4-Lipschitzness of sigmoid, $\|r\| \le \sqrt{n}$, and $\|w^t\| = O_P(\sqrt{m})$, and iterate the resulting linear inequality for fixed T.

A.7 The perpendicular RF matrix satisfies generalized invariance

We verify the matrix assumptions of [5, Proposition D.1(b2)]. The following section then embeds the physical GD iteration in the corresponding causal recursion.

Let $N/d \to \delta > 0$ be the total number of rows, $p = d - 1$, $X \in \mathbb{R}^{N\times p}$ have independent standard Gaussian entries, and $G \in \mathbb{R}^{m\times p}$ be an independent standard Gaussian matrix. Set

$$V = G/\sqrt{d}, \qquad A_0 = \frac{1}{\sqrt{m}}\phi(XV^\top), \qquad r = N/m \to \delta/\gamma. \tag{A.7.1}$$

For the almost-sure source-theorem application below, choose the standard product construction in which the full data-array sequence is independent of the full feature-array sequence, and different dimensions may be taken independent. This choice is harmless for the final in-probability conclusions, which depend only on the per-dimension laws. It avoids conditioning on an entire feature sequence that could encode data from another dimension under an adversarial cross-dimension coupling.

Theorem 5 (matrix hypothesis verification). The matrix A_0 has the rectangular generalized-invariance property used by Wang–Zhong–Fan, with the same limiting diagonal distribution as

$$A_G = \frac{aXV^\top + \sqrt{\nu}\,E}{\sqrt{m}}, \tag{A.7.2}$$

where E is an independent standard Gaussian $N \times m$ matrix. In particular:

1. The operator norm is eventually bounded by a deterministic constant.
2. The singular-value distribution has a deterministic compactly supported limit, identical to that of (A.7.2), and its second moment is positive.

3. For every fixed integer $k \geq 0$ and every $\epsilon > 0$, the following inequalities hold eventually almost surely (with positive powers used for the first two lines):

$$\max_i \left| ((A_0 A_0^\top)^k)_{ii} - \frac{1}{N} \operatorname{tr}(A_0 A_0^\top)^k \right| + \max_{i \neq j} |((A_0 A_0^\top)^k)_{ij}| \leq m^{-1/2+\epsilon}, \tag{A.7.3}$$

$$\max_i \left| ((A_0^\top A_0)^k)_{ii} - \frac{1}{m} \operatorname{tr}(A_0^\top A_0)^k \right| + \max_{i \neq j} |((A_0^\top A_0)^k)_{ij}| \leq m^{-1/2+\epsilon}, \tag{A.7.4}$$

$$\max_{i,j} |((A_0 A_0^\top)^k A_0)_{ij}| \leq m^{-1/2+\epsilon}. \tag{A.7.5}$$

4. It is invariant under independent signed permutations of its rows and columns.

Proof of the uniform entry estimates. For a matrix B, let $\mathcal{P}_R(B)$ denote its Frobenius-metric projection onto the closed convex set $\{C : \|C\|_{\mathrm{op}} \leq R\}$. This map clips singular values at R and is Frobenius-nonexpansive. It commutes with left and right orthogonal transformations.

Fix sufficiently large deterministic constants C_X, C_G, C_A, and define

$$\bar{X} = \mathcal{P}_{C_X\sqrt{d}}(X), \quad \bar{G} = \mathcal{P}_{C_G\sqrt{d}}(G), \quad B = \mathcal{P}_{C_A}\left(\frac{1}{\sqrt{m}} \phi(\bar{X}\bar{G}^\top/\sqrt{d}) \right). \tag{A.7.6}$$

The exponential norm bounds in Section A.2 show $B = A_0$ outside an event of probability at most Ce^{-cd}. Thus $B = A_0$ eventually almost surely. This explicit clipping step avoids the invalid inference that a derivative bound only on a nonconvex high-probability set is automatically a global Lipschitz bound.

For any two base inputs, the product inequality and the scalar Lipschitz property of tanh give

$$\|B(X,G) - B(X',G')\|_F \leq \frac{C}{\sqrt{m}} (\|X - X'\|_F + \|G - G'\|_F) \leq \frac{C'}{\sqrt{d}} \|(X,G) - (X',G')\|_F. \tag{A.7.7}$$

Because $\|B\|_{\mathrm{op}} \leq C_A$, each scalar entry of $(BB^\top)^k$, $(B^\top B)^k$, or $(BB^\top)^k B$ is a globally C_k-Lipschitz function of B in Frobenius norm. This follows by telescoping products; for example

$$\|(BB^\top)^k - (B'B'^\top)^k\|_F \leq 2kC_A^{2k-1} \|B - B'\|_F.$$

Consequently each such scalar entry is $C'_k/\sqrt{d}$-Lipschitz in the independent standard Gaussian base entries.

Oddness of tanh and equivariance of all clipping maps imply that B is invariant under row and column signed permutations. Therefore every offdiagonal entry of $(BB^\top)^k$ or $(B^\top B)^k$ has expectation zero; all diagonal entries in a given block have the same expectation; and all entries of $(BB^\top)^k B$ have expectation zero. Gaussian concentration and a union bound over $O(d^2)$ entries give, for any fixed $\eta > 0$, failure probability at most

$$C_k d^2 \exp(-c_k d^{2\eta})$$

for the assertion that every entry differs from its expectation by at most $d^{-1/2+\eta}$. This sequence is summable. The difference between a diagonal expectation and the realized average trace is bounded by the same maximum diagonal error. Choose $\eta < \epsilon$ and absorb fixed constants and dimension ratios. Borel–Cantelli and $B = A_0$ eventually prove (A.7.3)–(A.7.5), simultaneously for the countable collection of integer k and rational positive ϵ. Arbitrary positive ϵ follows by decreasing it to a rational number.

Proof of spectral matching. Let

$$\Sigma_V = \mathbb{E}_x[\phi(Vx)\phi(Vx)^\top], \qquad C_V = a^2 VV^\top + \nu I_m.$$

The Hermite covariance calculation strengthens to almost sure operator convergence here:

$$\|\Sigma_V - C_V\|_{\mathrm{op}} \to 0. \tag{A.7.8}$$

Indeed Gaussian norm and spherical-inner-product tails imply, eventually almost surely,

$$\max_j |\|f_j\|^2 - 1| = O(\sqrt{\log d/d}), \qquad \max_{j\neq k} \frac{|f_j^\top f_k|}{\|f_j\|\|f_k\|} = O(\sqrt{\log d/d}).$$

The offdiagonal higher-odd-Hermite remainder has operator norm at most its maximal absolute row sum, hence at most $Cm(\log d/d)^{3/2} = o(1)$. The diagonal remainder and linear Hermite coefficient converge uniformly, proving (A.7.8).

Condition on any realization of the sequence V satisfying these bounds. The Louart–Liao–Couillet spectral theorem applies to $\phi(XV^\top)$, since $V^\top$ is bounded in operator norm, tanh is Lipschitz, and all dimensions are proportional. It gives the same spectral deterministic equivalent as a Gaussian sample matrix with row covariance Σ_V. A Gaussian coupling and

$$\|\Sigma_V^{1/2} - C_V^{1/2}\|_{\mathrm{op}} \le \|\Sigma_V - C_V\|_{\mathrm{op}}^{1/2}$$

replace Σ_V by C_V without changing the limiting spectral law. Conditional on V, (A.7.2) has exactly Gaussian independent rows with covariance C_V/m. Fubini's theorem transfers this conditional almost-sure conclusion to the joint model.

The empirical eigenvalue law of $VV^\top$ tends to the Marchenko–Pastur law with aspect γ, so the empirical law of C_V tends to its image under $x \mapsto \nu + a^2 x$. The fixed-point description in the spectral theorem consequently has a deterministic limit. More explicitly, if H is that image law, the feature-side squared-singular law is characterized for $\zeta > 0$ by

$$\delta_\zeta = \int \frac{t}{\zeta + rt/(1+\delta_\zeta)}\, dH(t), \qquad m(-\zeta) = \int \frac{1}{\zeta + rt/(1+\delta_\zeta)}\, dH(t). \tag{A.7.9}$$

The positive solution is unique: divide its defining equation by $1+\delta_\zeta$ to obtain $\delta_\zeta/(1+\delta_\zeta) = \int t/[\zeta(1+\delta_\zeta) + rt]\, dH(t)$, whose left side is strictly increasing and right side decreasing. The bound $0 \le \delta_\zeta \le (\sup \operatorname{supp} H)/\zeta$ makes continuity under weak convergence of uniformly compactly supported covariance laws immediate by subsequences. The deterministic operator bounds turn weak spectral convergence into convergence of every fixed spectral moment. Also $m^{-1}\operatorname{tr}(A_0^\top A_0) \to rv > 0$. **Normalization for the AMP source:** its singular-value law D is padded to the row dimension N, whereas (A.7.9) is normalized to the feature dimension m. For $k \ge 1$, $\mathbb{E}[D^{2k}] = r^{-1}\int \lambda^k\, d\mu_{\mathrm{feature}}(\lambda)$; in particular $\mathbb{E}D^2 = v$, not rv. The aspect ratio is $r = N/m$. As an additional normalization check, $\mathbb{E}D^4 = v^2 + r(v^2 + a^4\gamma)$, so the singular-value law is nonconstant. This also supplies the positive-variance singular-law hypothesis used by the AMP degeneracy-removal argument.

Finally (A.7.2) is bi-orthogonally invariant: Gaussian left multiplication of X, E and Gaussian right multiplication of V, E preserve their joint law. The independently signed-permutation representation required by the generalized-invariance definition follows by taking an independent copy of A_0 and multiplying it by independent uniform signed permutations. The law is unchanged, and these permutations can be independent of any side information already independent of A_0. Items 1–4 are precisely the sufficient matrix conditions of the Proposition D.1(b), which identifies the limiting diagonal distribution. □

Theorem 5 does not infer dynamics from an empirical spectral law. It verifies the additional entrywise-polynomial and independence hypotheses needed by a dynamical universality theorem. Combined with Proposition 4, the residual issue is the finite-time algorithmic embedding, rather than an unchecked Gaussian-equivalence conjecture for the RF matrix.

A.8 Causal embedding of the rank-one logistic recursion in rectangular AMP

This section supplies the exact algorithmic embedding and the deterministic-overlap comparison. The Gaussian query/state recursion and positive-definiteness proof are in Appendix B. The present formulas specify their normalization and show that no extra universality for arbitrary empirical scalar coefficients is being assumed.

Use a proportional block of additional test inputs, for example $N = 2n$, and the deterministic row mask $M_i = 1$ for the n training rows and 0 for the test rows. Set

$$\pi = n/N = 1/2, \quad r = N/m \to 2\alpha/\gamma, \quad \lambda = c/\alpha, \quad s = a\sqrt{\gamma}, \quad v = \sqrt{d}\, t. \tag{A.8.1}$$

Here the letter v denotes the feature-side teacher vector only within this section; the scalar feature moment in earlier sections remains $\mathbb{E}\tanh(G)^2$. Write $w_t = \sqrt{m}\, b^t$. Proposition 4 reduces the normalized feature matrix to

$$D = A_0 + \frac{s}{m} z v^\top.$$

The small differences between (λ_d, s_d, r_d) and their limits are harmless by the same finite-time stability estimate. The exact resulting physical iteration is

$$q_t^{\mathrm{emp}} = m^{-1} v^\top w_t, \quad S_t = A_0 w_t + s q_t^{\mathrm{emp}} z, \quad r_t = M \odot [Y - \sigma(S_t)], \tag{A.8.2}$$

$$\rho_t^{\mathrm{emp}} = N^{-1} z^\top r_t, \qquad w_{t+1} = w_t + \lambda[A_0^\top r_t + s r \rho_t^{\mathrm{emp}} v], \qquad w_0 = 0. \tag{A.8.3}$$

The side vectors (M, z, Y) and v are independent of A_0. Their empirical laws converge to (M, Z, Y) and V, where $M \sim \mathrm{Bernoulli}(\pi)$ is independent of (Z, Y), Z, V are standard Gaussian, and $Y \mid Z \sim \mathrm{Bernoulli}(\sigma(\beta Z))$. Gaussian and bounded side variables have all moments. Polynomial density follows, for example, by splitting the two finite coordinates M, Y and using polynomial density under the Gaussian measure; their joint law is dominated by a constant times Gaussian measure times counting measure.

Let a_{th}, b_{th} be the deterministic rectangular-AMP correction coefficients prescribed by the row-normalized singular law in Theorem 5. The AMP has column fields Z_t^{A}, row fields Y_t^{A}, feature iterates Q_t, and residual iterates R_t:

$$Z_t^{\mathrm{A}} = A_0^\top R_t - \sum_{h<t} b_{th} Q_h, \qquad Y_t^{\mathrm{A}} = A_0 Q_t - \sum_{h \le t} a_{th} R_h. \tag{A.8.4}$$

Initialize $Q_0 = 0$ and $R_1 = M \odot (Y - 1/2)$. Define its nonlinearities recursively by

$$Q_t = Q_{t-1} + \lambda \left[Z_t^{\mathrm{A}} + \sum_{h<t} b_{th} Q_h + s r \rho_{t-1} V \right], \tag{A.8.5}$$

$$R_{t+1} = M \left[Y - \sigma\left(Y_t^{\mathrm{A}} + \sum_{h \le t} a_{th} R_h + s q_t Z \right) \right], \tag{A.8.6}$$

where

$$\rho_{t-1} = \mathbb{E}[Z R_t], \qquad q_t = \mathbb{E}[V Q_t] \tag{A.8.7}$$

are taken in the limiting AMP law. Each Q_h in (A.8.5) is already a deterministic function of the past column fields and feature-side information, and each R_h in (A.8.6) is already a deterministic function of past row fields and sample-side information. Thus these are legitimate memory-dependent coordinatewise nonlinearities.

Causality. Before the tth column update, the law of R_t and hence ρ_{t-1} are already known. The column field law and (A.8.5) then determine Q_t and q_t. Only after this does (A.8.6) define R_{t+1}. Consequently the overlap definitions introduce no circular fixed point at a time step. The source's AMP covariance and correction-coefficient prescriptions follow the same alternating order.

Regularity. At a fixed horizon, (A.8.5) is affine in its Gaussian-field arguments. Equation (A.8.6) is recursively globally Lipschitz in its Gaussian-field arguments because $0 \le M \le 1$ and $\|\sigma'\|_\infty \le 1/4$. To meet a theorem requiring definitions on the entire Euclidean side-information space, replace the formal arguments M, Y by their projections onto $[0,1]$; this changes no observed value and keeps the Lipschitz constants uniform in the side arguments. All functions have at most polynomial growth. The Bernoulli sampling threshold is not an update nonlinearity: the already generated label is supplied as side information.

Exact cancellation. Substituting (A.8.4) into (A.8.5) cancels all b_{th} corrections. Substituting (A.8.4) into (A.8.6) cancels all a_{th} corrections. Thus the finite-dimensional AMP trajectory is exactly logistic GD with the scalar overlaps frozen to (A.8.7). These identities are algebraic and do not use a limit.

Deterministic-overlap freezing. Suppose the source theorem is applied to this AMP, using the positive-definiteness result of Appendix B. Let $(\bar{w}_t, \bar{r}_t)$ denote its frozen-overlap physical iterates. Joint empirical W_2 convergence gives

$$\epsilon_t^q := |m^{-1} v^\top \bar{w}_t - q_t| \to 0, \qquad \epsilon_t^\rho := |N^{-1} z^\top \bar{r}_t - \rho_t| \to 0, \tag{A.8.8}$$

since both products are continuous with quadratic growth. Define

$$e_t = \|w_t - \bar{w}_t\|_2/\sqrt{m}, \quad h_t = \|r_t - \bar{r}_t\|_2/\sqrt{N}, \quad V_m = \|v\|_2/\sqrt{m}, \quad Z_N = \|z\|_2/\sqrt{N}.$$

On $\|A_0\|_{\mathrm{op}} \le C$, Cauchy–Schwarz and sigmoid's derivative bound give

$$h_t \le \frac{1}{4}\left[(C/\sqrt{r} + sZ_N V_m)\, e_t + sZ_N \epsilon_t^q\right], \tag{A.8.9}$$

$$e_{t+1} \le e_t + \lambda\left[(C\sqrt{r} + srV_m Z_N)\, h_t + srV_m \epsilon_t^\rho\right]. \tag{A.8.10}$$

We have $e_0 = 0$, $V_m, Z_N \to 1$, and finitely many times. Induction using (A.8.8) proves $\max_{t \le T}(e_t + h_t) \to 0$ in probability. Differences in dimension-dependent constants add only $o_P(1)$ terms. Therefore the adaptive-overlap physical iteration has exactly the same finite-time empirical limit as the frozen AMP.

This comparison is independent of whether the source's remark on empirical Onsager coefficients includes the rank-one overlaps: it proves the required replacement directly.

A.9 Assembled finite-horizon theorem and its boundary

The preceding matrix inputs, Appendix B, and Appendix C give the following assembly.

Theorem 6 (original Gaussian tanh GD, each fixed finite horizon). Fix $\alpha, \gamma, c \in (0, \infty)$, $0 \le \beta < \infty$, and a fixed integer T. Run the original model (A.1.1). Compute $q_{1:T}, \Sigma_T$ by the explicit recursion (B.16) in Appendix B, with $s = a\sqrt{\gamma}$, and put

$$A_t = sq_t, \qquad Q_{st} = s^2 q_s q_t + (\Sigma_T)_{st}.$$

Then Σ_T is positive definite. Conditional on the realized features and training sample, the joint fresh-test law converges weakly in probability to

$$\begin{gathered}(Z, S_1, \dots, S_T, Y), \qquad S_t = A_t Z + G_t, \\ G_{1:T} \sim N(0, \Sigma_T), \quad G \perp Z, \qquad Y \mid Z \sim \mathrm{Bernoulli}(\sigma(\beta Z)).\end{gathered} \tag{A.9.1}$$

For every $1 \le t \le T$, the conditional law of (Z, S_t, Y) in fact converges in total variation in probability. The exact, unbinned, model-conditional calibration error converges in probability and every finite L^p to

$$C_t = \mathbb{E}_{S\sim N(0,Q_{tt})}|g_t(S) - \sigma(S)|, \quad g_t(x) = \mathbb{E}_H \sigma\left(\beta\left[\frac{A_t}{Q_{tt}}x + \sqrt{\frac{(\Sigma_T)_{tt}}{Q_{tt}}}\,H\right]\right). \tag{A.9.2}$$

Its population logistic risk converges in L^1 to

$$R_t = \mathbb{E}_G \log\left(2\cosh\left(\frac{\sqrt{Q_{tt}}G}{2}\right)\right) - \kappa A_t, \qquad \kappa = \mathbb{E}[Z(\sigma(\beta Z) - 1/2)]. \tag{A.9.3}$$

All these convergences are uniform over the finite index set $\{1, \dots, T\}$. At time zero, $C_0 = 0$ and $R_0 = \log 2$ exactly.

Assembly proof. Theorem 5 and the side-information/regularity checks in Section A.8 verify WZF's matrix and algorithm assumptions. the positive-definiteness induction in Appendix B verifies all of its nonsingularity assumptions without perturbation, for every $c > 0$. The AMP embedding cancels the memory corrections exactly; (A.8.9)–(A.8.10) remove frozen-overlap dependence; Proposition 4 and finite-time operator stability restore the actual tanh features. On a mask-zero test row all residual iterates vanish, so the limiting score is exactly $sq_tZ + G_t$, proving (A.9.1) for appended test rows.

Conditional on the original training environment those appended rows are independent fresh inputs. Conditional LLNs for bounded observables therefore yield the claimed conditional test-law convergence. For the covariance identification, Gaussian concentration gives $\mathbb{E}_X|S_t(X)|^4 \le C\|F\|_{\mathrm{op}}^4\|b^t\|_2^4 = O_P(1)$; hence conditional variances of the appended empirical quadratic averages are $O_P(N^{-1})$. This identifies the realized-model covariance limit with Q, including teacher covariances A.

Theorem 1 supplies coefficient delocalization. Since Appendix B proves $(\Sigma_T)_{tt} > 0$, the deterministic smoothing theorem in Appendix C upgrades each conditional pair to total variation. Adding the fresh label is a common Markov kernel and contracts total variation. the signed-measure inequality in Appendix C then proves (A.9.2). Gaussian integration gives (A.9.3); uniform second-moment bounds for the scores imply uniform integrability of logistic losses and conditional risks, so the convergence is in L^1. Finite unions give the stated uniformity over $t \le T$. □

For clarity, $C_t > 0$ at every positive finite time. If $\beta = 0$, the regression curve is $1/2$ and the score is nondegenerate. If $\beta > 0$ and $A_t \le 0$, monotonicity prevents g_t from equaling the increasing sigmoid. If $A_t > 0$, dominated convergence gives

$$1 - g_t(x) \sim \exp\left[-\frac{\beta A_t}{Q_{tt}}x + \frac{\beta^2(\Sigma_T)_{tt}}{2Q_{tt}}\right] \quad (x \to \infty).$$

Equality with $1 - \sigma(x) \sim e^{-x}$ would require both $\beta A_t/Q_{tt} = 1$ and $(\Sigma_T)_{tt} = 0$, contradicting positive definiteness. Continuity and the everywhere-positive score density turn nonidentity into strictly positive calibration error.

For any training-dependent stopping time $\tau_d \in \{0, \dots, T\}$, the realized population risk and calibration are approximated, in probability, by R_{τ_d} and C_{τ_d}. If the deterministic finite-grid risk has a unique minimizer, every minimizer of the realized population risk on that grid converges to it in probability. Minimizing calibration alone includes the exact zero-score solution at time zero. No monotonicity of the risk or calibration curve is asserted.

This theorem concerns every fixed finite horizon, at arbitrary positive finite aspect ratios and step size. It does not, by itself, identify an infinite-time limit or optimize over an unbounded stopping horizon. Appendix D gives an additional all-time result under its explicitly stated sample-ratio and small-step assumptions; those hypotheses must remain visible whenever that extension is used.

Appendix B. Explicit causal state evolution

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

B.1 Model, dimensions, and the physical update

Let $X_i \sim N(0, I_d)$, let θ be a deterministic unit vector, and let

$$Y_i \mid X_i \sim \text{Bernoulli}(\sigma(\beta\theta^\top X_i)), \qquad \sigma(x) = (1+e^{-x})^{-1}, \quad 0 \le \beta < \infty.$$

There are n training examples and m independent random features $W_j \sim N(0, I_d/d)$, with

$$n/d \to \alpha > 0, \qquad m/d \to \gamma > 0, \qquad H_{ij} = \tanh(W_j^\top X_i).$$

Write

$$a = \mathbb{E}[G \tanh G], \qquad v = \mathbb{E}[\tanh^2 G], \qquad \nu = v - a^2 > 0, \quad G \sim N(0,1).$$

Strict positivity follows because $\tanh G$ is not a scalar multiple of G in Gaussian L^2.

In Appendix A's normalization, $w^t = \sqrt{m}\, b^t$, $A = H/\sqrt{m}$, and a mean-loss GD step of size cd gives

$$w^0 = 0, \qquad w^{t+1} = w^t + \lambda_d A^\top r^t, \quad r^t = M \odot (Y - \sigma(Aw^t)), \quad \lambda_d = cd/n \to \lambda = c/\alpha. \tag{B.1}$$

Here a proportional independent query block is appended to the rows, and $M_i = 1$ on training rows and $M_i = 0$ on query rows. If the total number of rows is N, set

$$N/d \to \delta > \alpha, \qquad r = N/m \to \delta/\gamma, \qquad p = n/N \to \alpha/\delta. \tag{B.2}$$

The common choice $N = 2n$ gives $p = 1/2$, $\delta = 2\alpha$. No result depends on that choice.

Rotate θ to the first coordinate. Let $z_i = \theta^\top X_i$ and $h_j = \sqrt{d}\,\theta^\top W_j$. Then z_i, h_j are standard normal, and Appendix A's decomposition is

$$A = A_0 + \frac{s_d}{m} z h^\top + E_d, \quad s_d = a\sqrt{m/d} \to s = a\sqrt{\gamma}, \quad \|E_d\|_{\text{op}} = o_{\mathbb{P}}(1). \tag{B.3}$$

The perpendicular matrix A_0 is independent of (z, Y, M, h). For the exact spike matrix $A_0 + s_d z h^\top/m$, the physical GD update is

$$Q_t = Q_{t-1} + \lambda_d \left[A_0^\top R_t + s_d r_d \ell_t^{(d)} h \right], \quad R_{t+1} = M \odot \left[Y - \sigma(A_0 Q_t + s_d q_t^{(d)} z) \right], \tag{B.4}$$

where $Q_t = w^t$, $R_t = r^{t-1}$,

$$q_t^{(d)} = m^{-1} h^\top Q_t, \qquad \ell_t^{(d)} = N^{-1} z^\top R_t, \qquad r_d = N/m.$$

In particular the column spike coefficient is $sr\ell_t$, not $s\ell_t$.

B.2 The RF rectangular cumulants have a closed form

Let μ_k be the limiting **row-normalized** moments

$$\mu_k = \lim N^{-1} \operatorname{tr}(A_0 A_0^\top)^k.$$

The squared singular-value law is padded with zeros to N values when necessary. the Gaussian comparison matrix in Appendix A has independent rows conditional on its feature covariance, whose limiting covariance law is

$$\mathcal{H} = \mathcal{L}(\nu + a^2 L), \qquad L \sim \mathrm{MP}_\gamma, \quad \mathbb{E}L = 1.$$

Let $M(z) = \sum_{k\geq 1} \mu_k z^k$ and $M_{\mathcal{H}}(z) = \sum_{k\geq 1}(\int t^k \, d\mathcal{H}(t))z^k$. The spectral fixed point in Appendix A, expanded at infinity, gives the formal-power-series identity

$$rM(z) = M_{\mathcal{H}}(rz(1 + M(z))). \tag{B.5}$$

For completeness, writing $m_F(-\zeta)$ for the feature-side Stieltjes transform and e_ζ for the auxiliary scalar in Appendix A, its fixed point implies $re_\zeta/(1 + e_\zeta) = 1 - \zeta m_F(-\zeta)$. Since the feature moment series is rM, substitution $z = -1/\zeta$ gives $e_\zeta = -M/(1 + M)$ and then (B.5).

The ordinary free R-transform of $\mathcal{H}$ is

$$R^{\mathrm{ord}}_{\mathcal{H}}(u) = \nu + \frac{a^2}{1 - a^2\gamma u}.$$

Equivalently,

$$M_{\mathcal{H}}(w) = w(1 + M_{\mathcal{H}}(w)) \left[\nu + \frac{a^2}{1 - a^2\gamma w(1 + M_{\mathcal{H}}(w))}\right].$$

Combine this identity with (B.5), and set $t = z(1 + M)(1 + rM)$. Since $r\gamma = \delta$, this yields

$$M(z) = \nu t + \frac{a^2 t}{1 - a^2\delta t}. \tag{B.6}$$

Fan's rectangular moment/cumulant identity is $M(z) = R_{\mathrm{rect}}(z(1 + M(z))(1 + rM(z)))$. Therefore

$$\boxed{R_{\mathrm{rect}}(t) = \nu t + \frac{a^2 t}{1 - a^2\delta t}, \qquad \kappa_2 = v, \quad \kappa_{2k} = a^{2k}\delta^{k-1} \ (k \geq 2).} \tag{B.7}$$

This is a formal-series identity; no convergence radius or small-step condition is needed. In particular

$$\mu_1 = v, \qquad \mu_2 = (1 + r)v^2 + \delta a^4 = v^2 + r(v^2 + a^4\gamma), \qquad \kappa_4 = \delta a^4. \tag{B.8}$$

B.3 Exact finite-dimensional memory and covariance formulas

Use R_t for the scalar row residual and Q_t for the scalar column iterate. The centered row and column Gaussian fields will be G_t and Z_t, respectively. Their covariance matrices are Σ_t and Ω_t. The teacher scalar is denoted $Z_\star$, and the independent column teacher coordinate is $H_\star$.

Define the $t \times t$ matrices

$$\Delta_t = (\mathbb{E}[R_i R_j])_{i,j\leq t}, \quad \Gamma_t = (\mathbb{E}[Q_i Q_j])_{i,j\leq t},$$

$$(\Phi_t)_{ij} = \mathbb{E}\left[\frac{\partial R_i}{\partial G_j}\right] \quad (j < i), \qquad (\Psi_t)_{ij} = \mathbb{E}\left[\frac{\partial Q_i}{\partial Z_j}\right] \quad (j \leq i), \tag{B.9}$$

and put zero in the remaining entries. Thus Φ_t is strictly lower triangular and Ψ_t is lower triangular. All derivatives hold the deterministic recursion coefficients fixed.

Our coefficient matrices $\mathsf{A}_t = (a_{ij})$ and $\mathsf{B}_t = (b_{ij})$ are **lower** triangular. They are the transposes of Fan's displayed upper-triangular matrices A_t, B_t. Set $h_0 = a^2\delta$. Then the explicit formulas are

$$\boxed{\begin{aligned}
\mathsf{A}_t &= \nu\Psi_t + a^2\Psi_t(I - h_0\Phi_t\Psi_t)^{-1},\\
\mathsf{B}_t &= r\nu\Phi_t + ra^2\Phi_t(I - h_0\Psi_t\Phi_t)^{-1},\\
\Sigma_t &= \nu\Gamma_t + a^2(I - h_0\Psi_t\Phi_t)^{-1}\left[\Gamma_t + h_0\Psi_t\Delta_t\Psi_t^\top\right](I - h_0\Psi_t\Phi_t)^{-\top},\\
\Omega_t &= r\nu\Delta_t + ra^2(I - h_0\Phi_t\Psi_t)^{-1}\left[\Delta_t + h_0\Phi_t\Gamma_t\Phi_t^\top\right](I - h_0\Phi_t\Psi_t)^{-\top}.
\end{aligned}} \tag{B.10}$$

Every inverse is a finite polynomial: $\Phi_t\Psi_t$ and $\Psi_t\Phi_t$ are strictly lower triangular, so $(I - h_0L)^{-1} = \sum_{j=0}^{t-1} h_0^j L^j$.

Here is an explicit check against the primary-source formulas. With $L = \Phi\Psi$ and $K = \Psi\Phi$, Fan's covariance coefficients, in our notation, are

$$\Theta^{(j)} = \sum_{i=0}^{j} L^i\Delta(L^\top)^{j-i} + \sum_{i=0}^{j-1} L^i\Phi\Gamma\Phi^\top(L^\top)^{j-1-i},$$

$$\Xi^{(j)} = \sum_{i=0}^{j} K^i\Gamma(K^\top)^{j-i} + \sum_{i=0}^{j-1} K^i\Psi\Delta\Psi^\top(K^\top)^{j-1-i}.$$

The second sums are empty for $j = 0$. His prescriptions give $\Sigma = \sum_{j\geq 0}\kappa_{2j+2}\Xi^{(j)}$, $\Omega = r\sum_{j\geq 0}\kappa_{2j+2}\Theta^{(j)}$, $\mathsf{A} = \sum_{j\geq 0}\kappa_{2j+2}\Psi(\Phi\Psi)^j$, and $\mathsf{B} = r\sum_{j\geq 0}\kappa_{2j+2}\Phi(\Psi\Phi)^j$. Inserting (B.7) and summing the two finite geometric series proves (B.10). In particular no additional factor of r belongs inside either covariance bracket.

B.4 Causal recursion and its implementation without circular definitions

Let

$$Z_\star, H_\star \sim N(0,1), \quad Y \mid Z_\star \sim \text{Bernoulli}(\sigma(\beta Z_\star)), \quad M \sim \text{Bernoulli}(p),$$

with M independent of $(Z_\star, Y)$ and $H_\star$ on the separate column probability space. Initialize

$$R_1 = M(Y - 1/2), \qquad Q_0 = 0, \qquad q_0 = 0.$$

For $t = 1, 2, \ldots$, carry out the following operations.

1. From the already defined row functions $R_{1:t}(G_{1:t-1}, Z_\star, Y, M)$ and Σ_{t-1}, compute Δ_t, Φ_t, and $\ell_t = \mathbb{E}[Z_\star R_t]$.
2. Embed the previous Ψ_{t-1}, Γ_{t-1} in $t \times t$ matrices with zero last row/column. Compute B_t, Ω_t from (B.10). Their values do not depend on the missing current column function: the last column of Φ_t is zero, so $\Phi_t\Psi_t$ and $\Phi_t\Gamma_t\Phi_t^\top$ only use previous column information.
3. Let $Z_{1:t} \sim N(0, \Omega_t)$ independently of $H_\star$, consistently with previous stages. Define

$$Q_t = Q_{t-1} + \lambda\left[Z_t + \sum_{j<t} b_{tj}Q_j + sr\ell_t H_\star\right], \qquad q_t = \mathbb{E}[H_\star Q_t]. \tag{B.11}$$

4. Compute the current derivative row of Ψ_t and Γ_t, then compute A_t, Σ_t using (B.10). Let $G_{1:t} \sim N(0, \Sigma_t)$ independently of $(Z_\star, Y, M)$. Define

$$S_t = G_t + \sum_{j\leq t} a_{tj}R_j + sq_tZ_\star, \qquad R_{t+1} = M(Y - \sigma(S_t)). \tag{B.12}$$

All column functions are linear. More precisely,

$$Q_{1:t} = \Psi_t Z_{1:t} + q_{1:t} H_\star, \qquad \Gamma_t = \Psi_t \Omega_t \Psi_t^\top + q_{1:t} q_{1:t}^\top, \tag{B.13}$$

and their coefficients can be calculated without integration:

$$(\Psi_t)_{tj} = (\Psi_{t-1})_{t-1,j} + \lambda \mathbf{1}_{\{j=t\}} + \lambda \sum_{k<t} b_{tk} (\Psi_{t-1})_{kj},$$

$$q_t = q_{t-1} + \lambda \sum_{j<t} b_{tj} q_j + \lambda s r \ell_t, \tag{B.14}$$

where nonexistent entries at $t = 1$ are zero. In particular every diagonal entry of Ψ_t equals λ.

For the row derivatives, differentiate the already explicit scalar recursion:

$$\frac{\partial R_{t+1}}{\partial G_j} = -M\sigma'(S_t) \left[\mathbf{1}_{\{j=t\}} + \sum_{k \le t} a_{tk} \frac{\partial R_k}{\partial G_j} \right], \quad j \le t. \tag{B.15}$$

Thus the remaining computations are finite-dimensional Gaussian integrals and a Bernoulli average; there is no unknown limiting quantity outside the recursion.

B.5 Query-block size cancels exactly

The apparent dependence on δ in (B.7)–(B.12) is only a bookkeeping choice. Let R_t^{tr} be the row recursion conditioned on $M = 1$. Define

$$\widehat{\Delta}_t = \alpha(\mathbb{E}[R_i^{\mathrm{tr}} R_j^{\mathrm{tr}}])_{i,j \le t}, \qquad \widehat{\Phi}_t = \alpha(\mathbb{E}[\partial_{G_j} R_i^{\mathrm{tr}}])_{i,j \le t}, \qquad \widehat{\ell}_t = \alpha \mathbb{E}[Z_\star R_t^{\mathrm{tr}}].$$

Since $R_t = 0$ when $M = 0$,

$$\widehat{\Delta}_t = \delta \Delta_t, \qquad \widehat{\Phi}_t = \delta \Phi_t, \qquad \widehat{\ell}_t = \delta \ell_t.$$

Define $T_t = (I - a^2 \Psi_t \widehat{\Phi}_t)^{-1}$ and $U_t = (I - a^2 \widehat{\Phi}_t \Psi_t)^{-1}$. Then (B.10) becomes

$$\boxed{\begin{aligned} \mathsf{A}_t &= \nu \Psi_t + a^2 \Psi_t U_t, \\ \mathsf{B}_t &= \gamma^{-1} \left[\nu \widehat{\Phi}_t + a^2 \widehat{\Phi}_t T_t \right], \\ \Sigma_t &= \nu \Gamma_t + a^2 T_t \left[\Gamma_t + a^2 \Psi_t \widehat{\Delta}_t \Psi_t^\top \right] T_t^\top, \\ \Omega_t &= \gamma^{-1} \left\{ \nu \widehat{\Delta}_t + a^2 U_t \left[\widehat{\Delta}_t + a^2 \widehat{\Phi}_t \Gamma_t \widehat{\Phi}_t^\top \right] U_t^\top \right\}. \end{aligned}} \tag{B.16}$$

Replace $s r \ell_t$ in (B.11) by $(s/\gamma)\widehat{\ell}_t$ and use $M = 1$ in the training scalar recursion. This is an entirely δ-free prescription, depending only on $(\alpha, \gamma, \beta, c, a, \nu)$.

On a query row $M = 0$, every R_j is zero. The physical test path is therefore

$$\boxed{S_t^{\mathrm{test}} = s q_t Z_\star + G_t, \qquad G_{1:T} \sim N(0, \Sigma_T) \text{ independently of } Z_\star.} \tag{B.17}$$

Its deterministic teacher/score overlaps are

$$\mathcal{A}_t = s q_t, \qquad \mathcal{Q}_{st} = s^2 q_s q_t + (\Sigma_T)_{st}. \tag{B.18}$$

Equation (B.16) proves algebraically that these quantities are independent of the number of appended queries.

B.6 Positive definiteness: no regularization or algorithm change

Suppose $c > 0$. Equations (B.10) imply

$$\Omega_t \succeq r\nu\Delta_t, \qquad \Sigma_t \succeq \nu\Gamma_t. \tag{B.19}$$

Initially $\Delta_1 = p/4 > 0$. If $\Delta_t \succ 0$, then $\Omega_t \succ 0$. Since Ψ_t is triangular with diagonal $\lambda > 0$, (B.13) gives $\Gamma_t \succ 0$, hence $\Sigma_t \succ 0$.

To continue the induction, suppose R_{t+1} were in the linear span of $R_1, \dots, R_t$ almost surely. Condition on $(Z_\star, Y, M, G_1, \dots, G_{t-1})$. On the event $M = 1$ of positive probability, all previous residuals are fixed. But the conditional variance of G_t is positive because $\Sigma_t \succ 0$, and

$$R_{t+1} = Y - \sigma\left(G_t + \sum_{j\le t} a_{tj}R_j + sq_tZ_\star\right)$$

is a strictly monotone nonconstant function of that conditional Gaussian. This is impossible. Therefore $\Delta_{t+1} \succ 0$. Induction establishes positive definiteness at every finite stage, including the uninformative-teacher case $\beta = 0$.

The case $c = 0$ is immediate from (B.1): every score is zero. It need not be passed through a theorem requiring nonsingular covariances.

B.7 Why this is ordinary GD, including adaptive overlaps

For any fixed deterministic arrays a_{tj}, b_{tj}, introduce fields

$$\mathbf{Z}_t = A_0^\top \mathbf{R}_t - \sum_{j<t} b_{tj}\mathbf{Q}_j, \qquad \mathbf{G}_t = A_0\mathbf{Q}_t - \sum_{j\le t} a_{tj}\mathbf{R}_j.$$

Using (B.11) and (B.12) coordinatewise adds these same terms back. With empirical teacher overlaps, the resulting algorithm is identically (B.4), at every finite dimension. This algebra does not impose an Onsager correction on GD.

For the theorem application, first use the deterministic scalar overlaps q_t, ℓ_t of (B.11)–(B.12). The resulting auxiliary AMP has coordinatewise nonlinearities, so WZF applies. Its empirical overlaps converge to q_t, ℓ_t by its joint W_2 convergence with the teacher side information. Then compare it with the empirical-overlap recursion (B.4).

Here are explicit bounds closing that comparison. Normalize errors by $e_t^Q = \|Q_t^{\mathrm{GD}} - Q_t^{\mathrm{aux}}\|/\sqrt{m}$ and $e_t^R = \|R_t^{\mathrm{GD}} - R_t^{\mathrm{aux}}\|/\sqrt{N}$. On events where $\|A_0\|_{\mathrm{op}}$, $\|h\|/\sqrt{m}$, $\|z\|/\sqrt{N}$, and the fixed dimension ratios are bounded, Cauchy–Schwarz gives

$$|q_t^{\mathrm{GD}} - q_t| \le Ce_t^Q + |m^{-1}h^\top Q_t^{\mathrm{aux}} - q_t|,$$

$$|\ell_t^{\mathrm{GD}} - \ell_t| \le Ce_t^R + |N^{-1}z^\top R_t^{\mathrm{aux}} - \ell_t|.$$

The two trailing terms tend to zero. The physical updates and $\mathrm{Lip}(\sigma) = 1/4$ then imply

$$e_t^Q \le e_{t-1}^Q + Ce_t^R + o_{\mathbb{P}}(1), \qquad e_{t+1}^R \le Ce_t^Q + o_{\mathbb{P}}(1). \tag{B.20}$$

The initial error is zero, so both errors vanish at every fixed time. Replacing s_d, λ_d, r_d by their limits only adds vanishing terms of the same kind.

Finally, $\|E_d\|_{\mathrm{op}} = o_{\mathbb{P}}(1)$ in (B.3) transfers this conclusion to the **actual tanh RF matrix**. At fixed time $\|Q_t\|/\sqrt{m} = O_{\mathbb{P}}(1)$ and $\|R_t\|/\sqrt{N} \le 1$. Thus every occurrence of E_dQ_t or $E_d^\top R_t$ is

$o_{\mathbb{P}}(1)$ in the relevant normalized norm, and the same finite induction proves convergence of actual-GD iterates and score vectors.

The appended queries never enter the update, since their mask is zero. Conditional on the training data and features, their score paths are iid fresh-test paths. A conditional law of large numbers for bounded Lipschitz functions therefore converts empirical query convergence into convergence in probability of the conditional fresh-test law to (B.17). Quadratic observables also converge: the fixed-time operator bounds and Gaussian Lipschitz concentration give bounded conditional fourth moments in probability, so the conditional variance of their query averages is $O_{\mathbb{P}}(1/N)$. Consequently the realized fresh-test covariance in Appendix A converges to (B.18).

B.8 Source hypotheses and normalization

The primary sources are:

- Zhou Fan, *Approximate Message Passing algorithms for rotationally invariant matrices*, Annals of Statistics 50(1), 197–224 (2022), arXiv:2008.11892v5. In that paper, Section 2.4 gives the rectangular cumulant convention, and Section 5.1, equations (5.7)–(5.13), gives the memory/covariance prescription. Corollary 5.4 addresses degeneracy for the invariant comparison ensemble.
- Tianhao Wang, Xinyi Zhong, and Zhou Fan, *Universality of Approximate Message Passing algorithms and tensor networks*, Annals of Applied Probability 34(4), 3943–3994 (2024), arXiv:2206.13037v5. The applicable result is Theorem 2.22 with Assumption 2.17, algorithm (2.14), and prescriptions (D.4)–(D.5). Proposition D.1(b) supplies the matrix criterion checked in Appendix A.

The application checks are as follows.

Requirement	Verification
Generalized-invariant matrix with invariant comparison law	Appendix A Theorem 5, applied to the perpendicular RF matrix with all training/query rows present
Independent initialization and side information	A_0 is independent of z, Y, M, h after teacher-coordinate separation
Nonzero initial AMP second moment	$R_1 = M(Y - 1/2)$ and $\mathbb{E}R_1^2 = p/4$; zero GD weights do not mean zero AMP initialization
All side-information moments and polynomial density	Gaussian teacher coordinates and bounded labels/masks have exponential moments; their polynomial algebra is dense in L^2
Coordinatewise continuous, polynomial-growth, Lipschitz functions	Column functions are linear; row functions are finite compositions of sigmoid and affine functions with deterministic coefficients
Uniform Lipschitz constants over side information	Extend the mask off $\{0, 1\}$ by the smooth bounded function $\sin^2(\pi x/2)$; this leaves actual iterates unchanged and keeps row derivatives uniformly bounded
Eventual operator bound and positive spectral second moment	Appendix A Theorem 5; $\mathbb{E}D^2 = v > 0$
Nonsingular Ω_t, Σ_t	The direct induction in Section B.6
Empirical teacher-overlap dependence	Apply the theorem to deterministic overlaps first, then use (B.20)

Requirement	Verification
Fixed time and consistent finite aspect ratios	Every statement fixes T before $d \to \infty$; deterministic parameter replacements are handled by (B.20)
Fresh-test extraction	Mask-zero appended rows, (B.17), and the conditional law of large numbers

The source notation uses a matrix with "m rows and n columns." Here these correspond to N rows and m features, so its aspect parameter is $r = \delta/\gamma$, **not** the RF ratio $\gamma = m/d$. The zero padding in Section B.2 is essential when $N > m$.

B.9 Exact first-step validation

Put

$$\mu = \mathbb{E}[Z_\star(\sigma(\beta Z_\star) - 1/2)].$$

Using the query-independent recursion (B.16),

$$\widehat{\Delta}_{11} = \alpha/4, \quad \widehat{\Phi}_{11} = 0, \quad \widehat{\ell}_1 = \alpha\mu, \quad \Omega_{11} = \frac{\alpha v}{4\gamma}, \quad \Psi_{11} = \lambda.$$

Consequently

$$Q_1 = \lambda Z_1 + \frac{ca\mu}{\sqrt{\gamma}} H_\star, \quad q_1 = \frac{ca\mu}{\sqrt{\gamma}}, \quad \Gamma_{11} = \frac{c^2 v}{4\alpha\gamma} + \frac{c^2 a^2 \mu^2}{\gamma}.$$

The forward memory coefficient is $a_{11} = \lambda v$. It vanishes from a test score because $R_1 = 0$ on the test block. The Gaussian residual variance is

$$\begin{aligned}\Sigma_{11} &= v\Gamma_{11} + a^4\lambda^2\widehat{\Delta}_{11} \\ &= c^2\left[\frac{\mu^2 a^2 v}{\gamma} + \frac{a^4 + v^2/\gamma}{4\alpha}\right].\end{aligned}$$

Since $sq_1 = ca^2\mu$, the fresh-test law is exactly

$$\boxed{S_1^{\text{test}} = ca^2\mu\, Z_\star + au_1 G, \qquad \tau_1^2 = c^2\left[\frac{\mu^2 a^2 v}{\gamma} + \frac{a^4 + v^2/\gamma}{4\alpha}\right],} \tag{B.21}$$

with G independent standard normal. This reproduces both terms of the first-step variance. Omitting $\kappa_4 = \delta a^4$ would incorrectly remove $c^2 a^4/(4\alpha)$; forgetting the mask-zero query block would incorrectly retain the training self term $\lambda v(Y - 1/2)$.

Appendix C. Total variation and exact calibration transfer

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

C.1 Notation and the exact calibration functional

Write

$$X \sim N(0, I_d), \quad Z = \theta_d^\top X, \quad \|\theta_d\| = 1, \quad S = f_d(X),$$
$$Y \mid X, \mathcal{E}_d \sim \text{Bernoulli}(p(Z)), \qquad p(z) = \sigma(\beta z).$$

The environment $\mathcal{E}_d$ contains the training data and random features, and the fresh X is independent of this environment. A realized score is a deterministic function of this fresh input. We use the **total-variation norm** convention

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{|h|\le 1} \left| \int h \, d\mu - \int h \, d\nu \right|, \tag{C.1.1}$$

which is twice $\sup_A |\mu(A) - \nu(A)|$ for probability measures.

Let μ be the conditional law of (Z, S). Its exact calibration functional is

$$\mathcal{C}_\beta(\mu) = \mathbb{E}_\mu \left| \mathbb{E}_\mu[p(Z) \mid S] - \sigma(S) \right|. \tag{C.1.2}$$

This equals the original calibration error conditioned on the predicted probability and the environment, since sigmoid is one-to-one and $\mathbb{E}[Y \mid X, \mathcal{E}_d] = p(Z)$.

For a finite signed measure on the score line define

$$\zeta_\mu(B) = \int \mathbf{1}_{\{s \in B\}} [p(z) - \sigma(s)] \, \mu(dz, ds).$$

Its Radon–Nikodym derivative relative to the score marginal is $\mathbb{E}_\mu[p(Z) \mid S = s] - \sigma(s)$, so

$$\mathcal{C}_\beta(\mu) = \|\zeta_\mu\|_{\mathrm{TV}}.$$

Consequently, for any two probability laws on $\mathbb{R}^2$,

$$\boxed{|\mathcal{C}_\beta(\mu) - \mathcal{C}_\beta(\nu)| \le \|\mu - \nu\|_{\mathrm{TV}}.} \tag{C.1.3}$$

Indeed $|p(z) - \sigma(s)| \le 1$, and multiplication by this bounded function followed by projection onto the score coordinate contracts the total-variation norm. Thus no continuity of a conditional-expectation operator under weak convergence is being asserted.

There is a useful stronger statement. Let $g_\mu(s) = \mathbb{E}_\mu[p(Z) \mid S = s]$, and choose an everywhere $[0, 1]$-valued version of g_ν on the whole score line. The same signed-measure argument, applied to the positive measures $g_\mu \mu_S$ and $g_\nu \nu_S$, gives

$$\int |g_\mu(s) - g_\nu(s)| \, \mu_S(ds) \le 2\|\mu - \nu\|_{\mathrm{TV}}. \tag{C.1.4}$$

This is convergence of the conditional calibration curve integrated under the actual score law. It does not assert pointwise convergence at every score.

C.2 A deterministic Gaussian smoothing theorem

All norms on derivatives in this section are uniform over $x \in \mathbb{R}^d$. A Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$.

Theorem 2.1 (joint TV approximation with matched covariance). Let $f_d : \mathbb{R}^d \to \mathbb{R}$ be smooth, odd functions, and let $X_d \sim N(0, I_d)$. Suppose, for fixed finite L, H,

$$\|\nabla f_d\| \le L, \qquad \|\nabla^2 f_d\|_{\mathrm{HS}} \le H, \qquad \|\nabla^2 f_d\|_{\mathrm{op}} \le \varepsilon_d \longrightarrow 0. \tag{C.2.1}$$

Put

$$A_d = \mathbb{E}[Z_d f_d(X_d)], \quad Q_d = \mathbb{E} f_d(X_d)^2, \quad \delta_d = Q_d - A_d^2, \quad Z_d = \theta_d^\top X_d.$$

If $\delta_d \ge \delta_* > 0$, then

$$\left\| \mathcal{L}(Z_d, f_d(X_d)) - N\left(0, \begin{pmatrix} 1 & A_d \\ A_d & Q_d \end{pmatrix}\right) \right\|_{\mathrm{TV}} \longrightarrow 0. \tag{C.2.2}$$

The covariance need not converge. In particular, if the pair converges weakly to a nonsingular centered Gaussian with covariance $\left(\begin{smallmatrix} 1 & A \\ A & Q \end{smallmatrix}\right)$, then its convergence is in total variation.

The proof has two separate parts: weak Gaussian approximation from the small Hessian operator norm, and a smoothing estimate using the bounded Hessian Hilbert–Schmidt norm. Supplying both is necessary for a joint TV conclusion.

C.2.1 Self-contained weak Gaussian approximation

For a centered smooth Gaussian functional $f(X)$, define the Ornstein–Uhlenbeck semigroup

$$P_t f(x) = \mathbb{E} f(e^{-t}x + \sqrt{1 - e^{-2t}} X'), \qquad U_f(x) = \int_0^\infty e^{-t} P_t \nabla f(x)\, dt.$$

Gaussian integration by parts and the semigroup generator give the covariance identity

$$\mathbb{E}[f(X)h(X)] = \mathbb{E}[U_f(X) \cdot \nabla h(X)]. \tag{C.2.3}$$

For completeness, $f - \mathbb{E}f = -\int_0^\infty \partial_t P_t f\, dt$. The generator is $\mathcal{L} = \Delta - x \cdot \nabla$, and $-\mathbb{E}[(\mathcal{L}g)h] = \mathbb{E}\nabla g \cdot \nabla h$. Also $\nabla P_t f = e^{-t} P_t \nabla f$. Substitution proves (C.2.3). The bounded derivative assumptions, Gaussian moments, and an initial truncation justify these integrations for the functions used here.

If $\|\nabla f\| \le L_f$ and $\|\nabla^2 f\|_{\mathrm{op}} \le \varepsilon_f$, then

$$\|U_f\| \le L_f, \qquad \|DU_f\|_{\mathrm{op}} = \left\| \int_0^\infty e^{-2t} P_t \nabla^2 f\, dt \right\|_{\mathrm{op}} \le \varepsilon_f / 2.$$

Set $T_f = \nabla f \cdot U_f$. Equation (C.2.3), with $h = f$, gives $\mathbb{E}T_f = \mathrm{Var}(f(X)) =: v_f$. Moreover

$$\|\nabla T_f\| \le \tfrac{3}{2} L_f \varepsilon_f, \qquad \mathrm{Var}(T_f) \le \tfrac{9}{4} L_f^2 \varepsilon_f^2 \tag{C.2.4}$$

by Gaussian Poincare. Apply (C.2.3) to $h = e^{itf}$. The characteristic function $\chi_f(t) = \mathbb{E}e^{itf}$ satisfies

$$\chi_f'(t) = -t v_f \chi_f(t) - t\mathbb{E}[(T_f - v_f)e^{itf}].$$

Solving this scalar differential equation and using (C.2.4) gives

$$|\chi_f(t) - e^{-v_f t^2/2}| \le \tfrac{3}{4} L_f \varepsilon_f t^2. \tag{C.2.5}$$

Apply (C.2.5) to $f = uZ_d + v f_d(X_d)$, whose gradient is bounded by $|u| + |v|L$ and Hessian operator norm by $|v|\varepsilon_d$. Along every subsequence where $(A_d, Q_d) \to (A, Q)$, it follows that $(Z_d, f_d(X_d))$ converges weakly to the Gaussian with that covariance. Such covariance-convergent subsequences exist because Poincare gives $Q_d \le L^2$, and $|A_d| \le L$.

This derivation uses no independence of feature coordinates and no assumed random-feature Gaussian equivalence.

C.2.2 A nondegenerate Gaussian derivative matrix with high probability

Temporarily suppress the dimension subscript. Write

$$g = \nabla f, \quad J = \nabla^2 f, \quad P = I - \theta\theta^\top, \quad q = Pg, \quad \Delta = \|q\|^2.$$

The determinant of the Gaussian derivative Gram matrix of (Z, f) is exactly Δ. Let $a = \theta \cdot g$. Gaussian integration by parts gives $\mathbb{E}a = A$, while Poincare gives

$$\mathrm{Var}(a) \le \mathbb{E}\|J\theta\|^2 \le \varepsilon^2.$$

Using Poincare again for f,

$$\mathbb{E}\Delta = \mathbb{E}\|g\|^2 - \mathbb{E}a^2 \ge Q - A^2 - \varepsilon^2. \tag{C.2.6}$$

Also $\nabla\Delta = 2Jq$, so

$$\mathrm{Var}(\Delta) \le 4L^2\varepsilon^2. \tag{C.2.7}$$

If $Q - A^2 \ge \delta_*$, equations (C.2.6)–(C.2.7) imply

$$\Pr\{\Delta \le \delta_*/4\} = o(1). \tag{C.2.8}$$

For example, when $\varepsilon^2 \le \delta_*/4$, the left side is at most $16L^2\varepsilon^2/\delta_*^2$, since $\mathbb{E}\Delta - \delta_*/4 \ge \delta_*/2$.

Thus one needs neither a pointwise lower bound on Δ nor inverse moments of Δ over its entire distribution.

C.2.3 Localized integration by parts and the smoothing estimate

Choose a fixed smooth $\chi : [0,\infty) \to [0,1]$ equal to zero on $[0,\delta_*/8]$ and to one on $[\delta_*/4,\infty)$. On $\Delta > 0$ define

$$v_2 = \frac{q}{\Delta}, \qquad v_1 = \theta - a v_2.$$

They obey

$$\theta \cdot v_1 = 1, \quad g \cdot v_1 = 0, \qquad \theta \cdot v_2 = 0, \quad g \cdot v_2 = 1. \tag{C.2.9}$$

Set $V_i = \chi(\Delta) v_i$, extended by zero on the region where the cutoff vanishes. These are globally smooth vector fields. On the support of the cutoff, $\Delta \geq \delta_*/8$. The bounds

$$\|v_2\| = \Delta^{-1/2}, \qquad \|Dv_2\|_{\mathrm{HS}} \leq \frac{\|J\|_{\mathrm{HS}}}{\Delta} + \frac{\|q\|\|\nabla\Delta\|}{\Delta^2} \leq \frac{3H}{\Delta}$$

and $|a| \leq L$, $\|\nabla a\| \leq H$, together with the cutoff derivative bound, imply

$$\|V_i\| + \|DV_i\|_{\mathrm{HS}} \leq C(L,H,\delta_*), \qquad i = 1,2. \tag{C.2.10}$$

The constant is independent of input dimension. No trace-of-Hessian bound is used here.

For a vector field V, its Gaussian divergence is $\mathfrak{d}(V) = X \cdot V - \operatorname{div} V$. Two Gaussian integrations by parts give

$$\mathbb{E}\mathfrak{d}(V)^2 = \mathbb{E}\|V\|^2 + \mathbb{E}\sum_{i,j}(\partial_i V_j)(\partial_j V_i) \leq \mathbb{E}\|V\|^2 + \mathbb{E}\|DV\|_{\mathrm{HS}}^2. \tag{C.2.11}$$

This identity can first be proved for smooth truncated fields and then extended by Gaussian Sobolev approximation; (C.2.10) supplies the required integrability. In particular its bound is dimension-free, although the two terms defining $\mathfrak{d}(V)$ separately need not be.

For $\psi \in C_c^\infty(\mathbb{R}^2)$, (C.2.9) and integration by parts yield

$$\mathbb{E}[\chi(\Delta)\,\partial_i\psi(Z,f)] = \mathbb{E}[\psi(Z,f)\mathfrak{d}(V_i)]. \tag{C.2.12}$$

Therefore the positive subprobability measure

$$\eta(B) = \mathbb{E}[\chi(\Delta)\mathbf{1}_{\{(Z,f)\in B\}}]$$

has distributional first derivatives with total-variation norms bounded by $C(L,H,\delta_*)$. More directly, integrate $\psi(x+h)-\psi(x)$ along the segment from x to $x+h$ and use (C.2.12). Taking the supremum over smooth bounded test functions gives

$$\|\tau_h\eta - \eta\|_{\mathrm{TV}} \leq C\|h\|. \tag{C.2.13}$$

The same supremum equals the total-variation norm for finite regular Borel measures; smooth approximation of continuous compactly supported tests justifies this step.

Let ρ_r be a nonnegative smooth probability mollifier on $\mathbb{R}^2$, supported in a ball of radius r. Integrating (C.2.13) against the mollifier and writing $\mu = \mathcal{L}(Z, f)$ gives

$$\begin{aligned} \|\mu - \mu * \rho_r\|_{\mathrm{TV}} &\le 2\|\mu - \eta\|_{\mathrm{TV}} + \|\eta - \eta * \rho_r\|_{\mathrm{TV}} \\ &\le 2\mathbb{E}[1 - \chi(\Delta)] + Cr = o(1) + Cr. \end{aligned} \tag{C.2.14}$$

This is the required uniform regularization estimate for the joint law.

C.2.4 Completing the TV proof

Pass to any subsequence along which $(A_d, Q_d) \to (A, Q)$. Its Gaussian covariance is nonsingular because $Q - A^2 \ge \delta_*$. Section C.2.1 gives $\mu_d \Rightarrow G_{A,Q}$. For fixed $r > 0$, convolution densities $\mu_d * \rho_r$ converge pointwise to $G_{A,Q} * \rho_r$, since the translated mollifier is a bounded continuous test function. All these nonnegative densities have mass one, so Scheffe's argument gives convergence in L^1, equivalently TV. The elementary argument is $\int |p_d - p| = 2 - 2\int \min(p_d, p) \to 0$, by dominated convergence for the minimum.

Use (C.2.14), this fixed-r convergence, and $\|G_{A,Q} - G_{A,Q} * \rho_r\|_{\mathrm{TV}} \to 0$ as $r \downarrow 0$. First taking the dimension limit and then $r \downarrow 0$ proves $\|\mu_d - G_{A,Q}\|_{\mathrm{TV}} \to 0$. Gaussian densities with converging nonsingular covariance matrices also converge pointwise and in L^1, so $\|G_{A_d,Q_d} - G_{A,Q}\|_{\mathrm{TV}} \to 0$. The triangle inequality proves (C.2.2) on this subsequence. Compactness of the covariance parameters then proves (C.2.2) for the whole sequence. This completes Theorem 2.1.

For the last assertion of the theorem, bounded gradients and oddness give the uniform fourth-moment bound $\mathbb{E}f_d^4 \le 5L^4$: apply Poincare to f_d^2 and use $\mathbb{E}f_d^2 \le L^2$. Hence a weak limit gives convergence of A_d, Q_d to A, Q, and $Q > A^2$ supplies $\delta_* > 0$ eventually.

C.3 Structural theorem for trained random-feature readouts

Let F_d have rows $W_j^\top$, and, conditional on an arbitrary environment containing (F_d, b_d), let

$$f_d(x) = \sum_{j=1}^{m} b_{d,j} \tanh(W_j^\top x). \tag{C.3.1}$$

There is **no independence assumption between the trained b_d and the features or training sample**. The fresh input remains independent of the environment.

Writing $J_2 = \sup_u |\tanh'' u| \le 2$, the exact derivative bounds are

$$\begin{aligned} \|\nabla f_d\| &\le \|F_d\|_{\mathrm{op}} \|b_d\|_2, \\ \|\nabla^2 f_d\|_{\mathrm{op}} &\le J_2 \|F_d\|_{\mathrm{op}}^2 \|b_d\|_\infty, \\ \|\nabla^2 f_d\|_{\mathrm{HS}} &\le J_2 \|F_d\|_{\mathrm{op}}^2 \|b_d\|_2. \end{aligned} \tag{C.3.2}$$

For the last inequality use $\nabla^2 f_d = F_d^\top \operatorname{diag}(b_{d,j} \tanh''(W_j^\top x)) F_d$ and $\|ABC\|_{\mathrm{HS}} \le \|A\|_{\mathrm{op}} \|B\|_{\mathrm{HS}} \|C\|_{\mathrm{op}}$.

Let

$$A_d = \mathbb{E}_X[Z f_d(X) \mid \mathcal{E}_d], \quad Q_d = \mathbb{E}_X[f_d(X)^2 \mid \mathcal{E}_d], \quad \delta_d = Q_d - A_d^2. \tag{C.3.3}$$

Corollary 3.1 (conditional matched-Gaussian TV). If

$$\|F_d\|_{\mathrm{op}} = O_P(1), \qquad \|b_d\|_2 = O_P(1), \qquad \|b_d\|_\infty \to 0 \text{ in probability}, \qquad \delta_d^{-1} = O_P(1), \tag{C.3.4}$$

then, writing $\mu_d = \mathcal{L}_X(Z, f_d(X) \mid \mathcal{E}_d)$,

$$\|\mu_d - G_{A_d,Q_d}\|_{\mathrm{TV}} \to 0 \quad \text{in probability and every finite } L^p. \tag{C.3.5}$$

Here $\delta_d^{-1} = O_P(1)$ includes positivity with probability tending to one; values on a vanishing exceptional event can be assigned arbitrarily. The L^p conclusion uses the bound of two for TV.

To justify the use of deterministic Theorem 2.1 with random environments, first restrict to events with fixed bounds on $\|F_d\|_{\mathrm{op}}, \|b_d\|_2, \delta_d^{-1}$. The deterministic convergence is uniform in the sequential sense on each such bounded class: if uniformity failed, choose one violating deterministic environment at each dimension and contradict Theorem 2.1. The $O_P(1)$ assumptions make the complementary probabilities arbitrarily small. Finally use $\|b_d\|_\infty \to 0$. No almost-sure boundedness of the sequence of random environments is assumed.

The same reasoning gives a direct transfer version: if the first three conditions in (C.3.4) hold and the conditional law of $(Z, f_d(X))$ converges weakly in probability to a fixed nonsingular Gaussian, then its convergence is in TV in probability. Uniform fourth moments on each bounded environment event give convergence of the covariances, as in the last paragraph of Section C.2.

C.3.1 RF covariance, relative delocalization, and nondegeneracy

For Gaussian rows $W_j \sim N(0, I_d/d)$, define

$$a_1 = \mathbb{E}[G \tanh G], \quad v = \mathbb{E} \tanh^2 G, \quad \nu = v - a_1^2 > 0.$$

Strict positivity holds by the strict Cauchy–Schwarz inequality, since $\tanh G$ is not a multiple of G almost surely. Put $q_j = \|W_j\|^2$, $D_A = \mathrm{diag}(\mathbb{E} \tanh'(\sqrt{q_j} G))$, and $\Sigma_F = \mathbb{E}_X[h(X)h(X)^\top]$. The exact first-Hermite decomposition and the operator bound in Appendix A give

$$\Sigma_F = D_A F F^\top D_A + D_R + E_F, \quad \|D_R - \nu I\|_{\mathrm{op}} \to 0, \quad \|E_F\|_{\mathrm{op}} \to 0 \tag{C.3.6}$$

in probability, when $m/d \to \gamma \in (0, \infty)$. Here $(D_R)_{jj} = \mathbb{E} \tanh^2(\sqrt{q_j} G) - q_j (D_A)_{jj}^2$. The off-diagonal remainder obeys the cubic Hermite bound, whose squared Frobenius expectation is $O(d^{-1})$; row-norm concentration handles the diagonal.

Gaussian integration by parts gives the exact covariance $A_d = \theta^\top F^\top D_A b_d$. Thus, uniformly over all readout vectors, including data-dependent vectors,

$$\begin{aligned} Q_d - A_d^2 &= \|(I - \theta\theta^\top) F^\top D_A b_d\|^2 + b_d^\top (D_R + E_F) b_d \\ &\geq (\nu - o_P(1)) \|b_d\|_2^2. \end{aligned} \tag{C.3.7}$$

In particular, a norm bounded away from zero supplies the last condition of (C.3.4). There is also a scale-free alternative: if

$$\Pr\{b_d = 0\} \to 0, \qquad \frac{\|b_d\|_\infty}{\|b_d\|_2} \to 0 \quad \text{in probability}, \tag{C.3.8}$$

retain $\|F_d\|_{\mathrm{op}} = O_P(1)$, which the Gaussian-row hypothesis supplies, and normalize the score by $\|b_d\|_2$. Its coefficient norm is one and (C.3.7) gives a residual variance at least $\nu/2$ with probability tending to one. Corollary 3.1 applies to the normalized score. The invertible, environment-dependent rescaling of its second coordinate preserves TV, proving (C.3.5) for the original score as well. This variant does not require its raw variance to stay bounded away from zero.

Absolute coefficient delocalization alone cannot be silently substituted for relative delocalization when the whole coefficient vector vanishes. At exactly zero score the conditioning sigma-field changes, so this is a substantive distinction for calibration.

C.4 Later fixed-time readouts: exact Gaussian calibration proxy

For $Q > A^2$, define the deterministic scalar functional

$$C_\beta(A, Q) = \mathbb{E}_T \left| \mathbb{E}_H \sigma\left(\beta \left[\frac{A}{\sqrt{Q}} T + \sqrt{1 - \frac{A^2}{Q}}\, H \right] \right) - \sigma(\sqrt{Q}\, T) \right|, \tag{C.4.1}$$

where T, H are independent standard normals. Corollary 3.1 and (C.1.3) immediately imply

$$\mathrm{CE}_d - C_\beta(A_d, Q_d) \longrightarrow 0 \quad \text{in probability and every finite } L^p. \tag{C.4.2}$$

This is an exact calibration approximation for the original trained network with its own conditional covariance, not calibration under a different training distribution. It holds for any fixed training time whose coefficients satisfy (C.3.4), or the relative variant (C.3.8).

If additionally $A_d \to A$ and $Q_d \to Q > A^2$ in probability, bounded convergence in (C.4.1) gives the deterministic exact limit $\mathrm{CE}_d \to C_\beta(A, Q)$. This explicitly identifies what a later-time overlap or DMFT theorem must provide.

Appendix A proves fixed-horizon coefficient bounds and delocalization for actual GD. Appendix B proves positive residual variance at each fixed positive time for every fixed positive step and supplies the deterministic overlap limits. These results provide all inputs to the transfer theorem above.

C.4.1 Risk also has a covariance proxy

On events with bounded $\|F\|_{\mathrm{op}}\|b_d\|_2$, Gaussian Poincare and oddness bound the score second moment uniformly. Truncate logistic loss, use (C.3.5) for the bounded truncation, then let the truncation level grow. The upper bound $0 \le \ell(Y, S) \le \log 2 + |S|$ makes the tail uniformly small on these events. Localization therefore proves

$$L_d - \left\{ \mathbb{E}_T \log\bigl(2\cosh(\sqrt{Q_d} T/2)\bigr) - \kappa A_d \right\} \longrightarrow 0 \quad \text{in probability.} \tag{C.4.3}$$

The Gaussian formula uses $\mathbb{E}[S\sigma(\beta Z)] = A_d \mathbb{E}[Z\sigma(\beta Z)] = \kappa A_d$. An L^1 conclusion for (C.4.3) needs unconditional uniform integrability; it is not inferred from $O_P(1)$ bounds alone.

C.4.2 Positive calibration without deterministic overlap convergence

For every $\beta \geq 0$, $Q > A^2$, and $Q > 0$,

$$C_\beta(A, Q) > 0. \tag{C.4.4}$$

If $\beta = 0$ or $A = 0$, the conditional teacher mean is $1/2$, while $\sigma(S)$ is nonconstant. If $\beta > 0, A < 0$, the former mean is strictly decreasing in the score whereas the prediction is strictly increasing. If $\beta > 0, A > 0$, write the former mean as

$$g(s) = \mathbb{E}_H \sigma(\lambda s + rH), \quad \lambda = \beta A/Q > 0, \quad r = \beta\sqrt{1 - A^2/Q} > 0.$$

If calibration held, continuity and positivity of the Gaussian density would imply $g(s) = \sigma(s)$ for every real s. But dominated convergence gives

$$e^{\lambda s}(1 - g(s)) \longrightarrow e^{r^2/2} \quad (s \to +\infty).$$

Comparison with the logistic tail first forces $\lambda = 1$, then $e^{r^2/2} = 1$, a contradiction.

The map C_β is continuous on the region $Q > A^2$. Hence for each fixed $0 < \delta_* \leq Q_* < \infty$,

$$c_*(\beta, \delta_*, Q_*) := \min_{\substack{Q \leq Q_* \\ Q - A^2 \geq \delta_*}} C_\beta(A, Q) > 0. \tag{C.4.5}$$

The set is compact and $Q \geq \delta_*$. Therefore if a finite collection of trained times satisfies common deterministic upper and lower covariance bounds with probability tending to one, as well as the hypotheses of Corollary 3.1, then

$$\Pr\left\{ \min_{1 \leq t \leq T} \mathrm{CE}_{t,d} \geq c_*/2 \right\} \longrightarrow 1. \tag{C.4.6}$$

A union bound over the fixed number of times suffices. Deterministic overlap convergence is unnecessary for this qualitative conclusion. It is still necessary for an explicit deterministic trajectory of risk and calibration, unless another argument identifies those functionals without identifying the overlaps.

Appendix D. Uniform time control in the sample-rich regime

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

D.1 Model, constants, and theorem

Let

$$X_i \overset{\text{iid}}{\sim} N(0, I_d), \qquad Y_i \mid X_i \sim \text{Bernoulli}(\sigma(\beta\theta^\top X_i)), \quad \|\theta\| = 1,$$

with conditionally independent labels, fixed finite $\beta \geq 0$, and $\sigma(u) = (1+e^{-u})^{-1}$. Independently, the rows of $F \in \mathbb{R}^{m\times d}$ are $W_j \sim N(0, I_d/d)$. Put

$$h(x) = \tanh(Fx), \quad H_{ij} = h_j(X_i), \quad \mathcal{L}_n(b) = \frac{1}{n}\sum_{i=1}^{n} \{\log(1 + e^{h(X_i)^\top b}) - Y_i h(X_i)^\top b\}. \tag{D.1.1}$$

Assume $n/d \to \alpha > 0$, $m/d \to \gamma > 0$. The raw score is $b^\top h(x)$; in the alternative network notation $b = a/\sqrt{m}$. The GD update in the other notes is exactly

$$b^0 = 0, \qquad b^{t+1} = b^t - \eta_d \nabla \mathcal{L}_n(b^t), \qquad \eta_d = \frac{cd}{m}. \tag{D.1.2}$$

Here are explicit, deliberately conservative constants. For $G \sim N(0,1)$, define

$$a_1 = \mathbb{E}[G \tanh G], \quad v = \mathbb{E}\tanh^2 G, \quad \nu = v - a_1^2 > 0.$$

Choose any deterministic L for which $\|F\|_{\text{op}} \leq L$ with probability tending to one. The elementary Gaussian net bound permits, for example,

$$L = 8\sqrt{1+\gamma}.$$

Set

$$s = \nu/2, \qquad \kappa = \sqrt{s/2}, \qquad p_0 = \frac{s^2}{20L^4}, \qquad B = \sqrt{2\log(4/p_0)},$$

$$q_\beta = \sigma(-\beta B), \quad p_1 = q_\beta p_0/2, \quad \mathfrak{a} = \kappa p_1/4, \quad C_* = \max\left\{1, \left(\frac{32L}{\kappa p_1}\right)^2\right\}. \tag{D.1.3}$$

The ratio and step assumptions for this appendix are

$$\boxed{\alpha/\gamma > C_*, \qquad 0 < c < \frac{\gamma}{2\Lambda}, \quad \Lambda = 16L^2.} \tag{D.1.4}$$

These define a nonempty proportional regime for each fixed β, γ. No assertion that these thresholds are sharp is made.

Further set

$$R_0 = \frac{\log 2}{\mathfrak{a}}, \quad R = 2R_0, \quad C_H = 64L^2, \quad \delta = p_0/4, \quad M = R\sqrt{C_H/\delta},$$

$$\chi = \frac{\kappa^2 p_0}{16}, \qquad \mu = \chi\sigma'(M) > 0, \qquad \eta_- = \frac{c}{2\gamma}, \qquad q = 1 - \eta_-\mu \in (0,1). \tag{D.1.5}$$

All these constants are independent of d, although some are very small or very large.

Theorem 1 (whole GD trajectory). Under (D.1.4), with probability tending to one:

1. $\mathcal{L}_n(b) \geq \mathfrak{a}\|b\|$ for every $b \in \mathbb{R}^m$. There is a unique finite unregularized MLE b^*, with $\|b^*\| \leq R_0$.
2. Every GD iterate and every feature-deleted GD iterate defined in Section D.6 lies in $\{\|b\| \leq R_0\}$. On the larger ball $\{\|b\| \leq R\}$,
$$\mu I \preceq \nabla^2 \mathcal{L}_n(b) \preceq \Lambda I. \tag{D.1.6}$$
3. For every integer $t \geq 0$, simultaneously,
$$\|b^t - b^*\| \leq R_0 q^t. \tag{D.1.7}$$
4. The MLE and the whole trajectory are delocalized:
$$\|b^*\|_\infty + \sup_{t \geq 0} \|b^t\|_\infty = O_P\left(\sqrt{\frac{\log d}{d}}\right). \tag{D.1.8}$$
5. For a deterministic $r_* > 0$,
$$\inf_{t \geq 1} \|b^t\| \geq r_*, \qquad \|b^*\| \geq r_* \tag{D.1.9}$$
with probability tending to one. One permissible choice is
$$r_* = \frac{c}{4}\sqrt{\frac{v\rho_\beta}{\alpha\gamma}}, \qquad \rho_\beta = \mathbb{E}\sigma'(\beta G) > 0. \tag{D.1.10}$$

The label mechanism is not independent of the feature vector after conditioning on F. The proof below does not make that assumption.

D.2 The good feature event

Write $\Sigma_F = \mathbb{E}_X h(X)h(X)^\top$ and $g_F = \mathbb{E}_X[(\theta^\top X)h(X)]$. There is an event $\mathcal{F}_d$, depending only on F, on which
$$\|F\|_{\mathrm{op}} \leq L, \qquad \Sigma_F \succeq sI, \qquad \Sigma_F - g_F g_F^\top \succeq sI, \tag{D.2.1}$$
with $\mathbb{P}(\mathcal{F}_d^c) = O(d^{-1}) + O(de^{-c_F d})$ for some fixed $c_F > 0$. Constants in probability estimates may depend on fixed model parameters.

For completeness, the covariance claim follows directly from Hermite expansion. Put $q_j = \|W_j\|^2$, $A(q) = \mathbb{E}\tanh'(\sqrt{q}G)$, and $D_A = \mathrm{diag}(A(q_j))$. Oddness gives only odd Hermite coefficients, and Gaussian integration by parts identifies the linear coefficient. Thus
$$\Sigma_F = D_A F F^\top D_A + D_\nu + E, \quad (D_\nu)_{jj} = \mathbb{E}\tanh^2(\sqrt{q_j}G) - q_j A(q_j)^2, \tag{D.2.2}$$
where $E_{jj} = 0$ and, for $j \neq k$,
$$|E_{jk}| \leq |\rho_{jk}|^3, \qquad \rho_{jk} = \frac{W_j^\top W_k}{\|W_j\|\|W_k\|}. \tag{D.2.3}$$
Indeed, Cauchy–Schwarz on the Hermite coefficients of degree at least three bounds their sum by $|\rho_{jk}|^3$, since tanh has second moment at most one. Independent Gaussian directions give
$$\mathbb{E}\|E\|_F^2 \leq \frac{15m(m-1)}{d(d+2)(d+4)} = O(d^{-1}). \tag{D.2.4}$$
Uniform concentration of q_j around one and continuity give $\min_j (D_\nu)_{jj} \geq 3\nu/4$ except on an exponentially unlikely event. Markov's inequality gives $\|E\|_{\mathrm{op}} \leq \nu/4$ with failure $O(d^{-1})$. Also
$$g_F = D_A F\theta, \quad \Sigma_F - g_F g_F^\top = D_A F(I - \theta\theta^\top)F^\top D_A + D_\nu + E \succeq sI. \tag{D.2.5}$$

The operator norm bound follows by a Gaussian sphere-net argument. These observations prove (D.2.1).

Conditional on $F \in \mathcal{F}_d$, for every unit $u \in \mathbb{R}^m$, the function $x \mapsto u^\top h(x)$ is odd and L-Lipschitz. In particular,

$$\mathbb{E}e^{tu^\top h(X)} \le e^{t^2L^2/2}, \qquad s \le \mathbb{E}(u^\top h(X))^2 \le L^2, \quad \mathbb{E}(u^\top h(X))^4 \le 5L^4. \tag{D.2.6}$$

The fourth-moment bound follows from Gaussian Poincare applied first to the centered function and then to its square: $\mathbb{E}f^4 \le (\mathbb{E}f^2)^2 + 4L^2\mathbb{E}f^2 \le 5L^4$.

D.3 Uniform small-ball events and noisy-label coercivity

Everything in this section is conditional on a fixed $F \in \mathcal{F}_d$. The pairs (X_i, Y_i) are iid under this conditioning. For each unit u, Paley–Zygmund, or Cauchy–Schwarz applied above the threshold, gives

$$\mathbb{P}(|u^\top h(X)| \ge \kappa) \ge p_0. \tag{D.3.1}$$

Let $Z = \theta^\top X$ and $T = 2Y - 1$. The chosen B has $\mathbb{P}(|Z| > B) \le p_0/2$. On $|Z| \le B$, either label has conditional probability at least q_β. Therefore

$$\mathbb{P}(-Tu^\top h(X) \ge \kappa) \ge q_\beta \mathbb{P}(|u^\top h(X)| \ge \kappa, |Z| \le B) \ge p_1. \tag{D.3.2}$$

This is a conditional-label calculation at each X; it never separates the dependent events involving $h(X)$ and Z.

Define the bounded cutoff

$$\psi(x) = \begin{cases} 0, & x \le \kappa/2, \\ 2x/\kappa - 1, & \kappa/2 < x < \kappa, \\ 1, & x \ge \kappa. \end{cases} \tag{D.3.3}$$

It is $2/\kappa$-Lipschitz and vanishes at zero. For either class

$$f_u(X, Y) = \psi(-Tu^\top h(X)), \qquad \widetilde{f}_u(X) = \psi(|u^\top h(X)|), \quad \|u\| = 1,$$

symmetrization and Rademacher contraction give

$$\mathbb{E} \sup_{\|u\|=1} |(P_n - P)f_u| \le \frac{8}{\kappa}\mathbb{E}\left\|\frac{1}{n}\sum_i \epsilon_i T_i h(X_i)\right\| \le \frac{8L}{\kappa}\sqrt{\frac{m}{n}}. \tag{D.3.4}$$

The same bound holds for $\widetilde{f}_u$, with T_i omitted. The numerical factor eight safely includes the absolute-value form of contraction. The last inequality uses $T_i^2 = 1$ and the independent Rademacher signs, not independence of T_i and $h(X_i)$.

Since changing one sample changes each supremum by at most $1/n$, bounded differences imply an upper-tail bound e^{-2nz^2} above its expectation. If $n/m \ge C_*$, (D.3.4) is at most $p_1/4$. Using $z = p_1/4$ for the signed class and $z = p_0/4$ for the unsigned class proves, simultaneously, except with probability

$$e^{-np_1^2/8} + e^{-np_0^2/8}, \tag{D.3.5}$$

that for every unit u,

$$\frac{1}{n}\sum_i \psi(-T_i u^\top h_i) \ge p_1/2, \qquad \frac{1}{n}\#\{i : |u^\top h_i| > \kappa/2\} \ge p_0/2. \tag{D.3.6}$$

Measurability of the suprema is unproblematic: the cutoff classes are continuous in u, so a countable dense subset of the sphere suffices.

The logistic loss obeys $\log(1+e^{-Tz}) \geq (-Tz)_+$. Also $x_+ \geq (\kappa/2)\psi(x)$. Consequently, for every $b \neq 0$, using $u = b/\|b\|$,

$$\mathcal{L}_n(b) \geq \|b\| \frac{1}{n} \sum_i (-T_i u^\top h_i)_+ \geq \frac{\kappa p_1}{4} \|b\| = \mathfrak{a}\|b\|. \tag{D.3.7}$$

The inequality at $b = 0$ is immediate. Thus the entire initial sublevel set $\{\mathcal{L}_n \leq \log 2\}$ is contained in the deterministic radius-R_0 ball.

D.4 Robust covariance and curvature on the whole ball

The unsigned event in (D.3.6) gives an arbitrary-deletion bound: for **every** subset $I \subseteq \{1, \ldots, n\}$ with $|I^c| \leq \delta n$,

$$\frac{1}{n} \sum_{i \in I} h_i h_i^\top \succeq \chi I_m. \tag{D.4.1}$$

Indeed, for each unit u, at least $p_0 n/2$ rows have $|u^\top h_i| > \kappa/2$; after arbitrary deletion of at most $p_0 n/4$ rows, at least $p_0 n/4$ remain, contributing at least $\kappa^2 p_0/16$. This proof does not union-bound over subsets and does not assume deleted rows are independent of the data.

We also have, with conditional failure at most $2e^{-3n}$ when $n \geq m$,

$$\|H\|_{\mathrm{op}} \leq 8L\sqrt{n}, \qquad \frac{1}{n} H^\top H \preceq C_H I_m. \tag{D.4.2}$$

For a direct proof, take 1/4-nets of the two unit spheres, of cardinalities at most $9^n, 9^m$. Each bilinear form $u^\top H v$ is centered L-subGaussian by independence of the rows and (D.2.6). The bound at $4L\sqrt{n}$, a union bound, and the factor-two net inequality give failure $2\exp\{(n+m)\log 9 - 8n\} \leq 2e^{-3n}$. This uses no within-row independence.

For any $\|b\| \leq R$, let $I_b = \{i : |h_i^\top b| \leq M\}$. Equation (D.4.2) gives

$$\frac{|I_b^c|}{n} \leq \frac{b^\top H^\top H b}{nM^2} \leq \frac{C_H R^2}{M^2} = \delta.$$

The logistic Hessian is therefore bounded simultaneously on the ball by

$$\nabla^2 \mathcal{L}_n(b) = \frac{1}{n} \sum_i \sigma'(h_i^\top b) h_i h_i^\top \succeq \sigma'(M) \frac{1}{n} \sum_{i \in I_b} h_i h_i^\top \succeq \mu I_m. \tag{D.4.3}$$

Globally, $\nabla^2 \mathcal{L}_n(b) \preceq C_H I/4 = \Lambda I$. Thus the desired curvature is simultaneous for all points in the ball, not merely at a prespecified iterate.

Let $\mathcal{G}_d$ be the feature event together with (D.3.6) and (D.4.2). For large d, (D.1.4) gives $n/m \geq C_*$, and

$$\mathbb{P}(\mathcal{G}_d^c) \leq \mathbb{P}(\mathcal{F}_d^c) + e^{-np_1^2/8} + e^{-np_0^2/8} + 2e^{-3n} = o(1). \tag{D.4.4}$$

D.5 MLE existence and uniform contraction

On $\mathcal{G}_d$, coercivity and continuity give a finite global minimizer. Its loss is at most $\mathcal{L}_n(0) = \log 2$, so its norm is at most R_0. Strict convexity on the containing ball proves uniqueness; call the minimizer b^*. Its full gradient is zero.

For probabilistic expressions involving the MLE, assign $b^* = 0$ on any exceptional realization without a unique finite MLE. This arbitrary extension has vanishing probability and does not change any conclusion.

For large d, the step assumption implies

$$\eta_- \le \eta_d \le 2c/\gamma < 1/\Lambda. \tag{D.5.1}$$

The global smoothness inequality gives

$$\mathcal{L}_n(b - \eta_d \nabla \mathcal{L}_n(b)) \le \mathcal{L}_n(b) - \frac{\eta_d}{2} \|\nabla \mathcal{L}_n(b)\|^2. \tag{D.5.2}$$

Thus all iterates remain in the initial sublevel set, and hence in the radius-R_0 ball.

For x, y in this ball, integrate the Hessian along their line segment. It yields a symmetric matrix $A_{x,y}$ with spectrum in $[\mu, \Lambda]$ such that

$$[x - \eta_d \nabla \mathcal{L}_n(x)] - [y - \eta_d \nabla \mathcal{L}_n(y)] = (I - \eta_d A_{x,y})(x - y).$$

Consequently the GD map contracts by

$$q_d = 1 - \eta_d \mu \le q < 1. \tag{D.5.3}$$

Applying this with $y = b^*$ proves (D.1.7), uniformly for every integer time. No dimension-dependent burn-in is hidden in the contraction factor.

D.6 Leave-one-feature definitions and concentration

Let P_j set coordinate j to zero. Define restricted GD for every data realization, including outside $\mathcal{G}_d$, by

$$b^{-j,0} = 0, \qquad b^{-j,t+1} = b^{-j,t} - \eta_d P_j \nabla \mathcal{L}_n(b^{-j,t}). \tag{D.6.1}$$

The coordinate j stays zero, so this update uses only X, Y, F_{-j}. On $\mathcal{G}_d$, the same descent inequality applied on the coordinate subspace shows that every restricted iterate has loss at most $\log 2$ and norm at most R_0. Therefore the same full Hessian bounds apply simultaneously along the full and all restricted trajectories.

Also define $\bar{b}^{-j}$, for every data realization, to minimize $\mathcal{L}_n$ over the deterministic compact set

$$\{b : b_j = 0,\ \|b\| \le R\}. \tag{D.6.2}$$

If needed, select the unique minimum-norm minimizer. This is a measurable selector: it is the limit as $\epsilon \downarrow 0$ of the unique minimizers after adding $\epsilon\|b\|^2$. The definition uses only X, Y, F_{-j}. On $\mathcal{G}_d$, its loss is at most $\log 2$, so $\|\bar{b}^{-j}\| \le R_0 < R$. It is then an unconstrained minimizer in its coordinate subspace, and

$$\nabla \mathcal{L}_n(\bar{b}^{-j}) = g_j^* e_j. \tag{D.6.3}$$

The larger compact ball in the definition is essential: it keeps the object well defined and independent of W_j even on exceptional samples, while still giving interior stationarity on the good event.

For any of these restricted coefficient vectors b, put

$$g_j(b) = \frac{1}{n} \sum_i [\sigma(h_{i,-j}^\top b_{-j}) - Y_i] \tanh(W_j^\top X_i). \tag{D.6.4}$$

Conditional on X, Y, F_{-j}, the bracketed residual vector is fixed, has norm at most $\sqrt{n}$, and is independent of $W_j = G_j/\sqrt{d}$. As a function of standard Gaussian G_j, (D.6.4) is odd, has mean zero, and has Lipschitz constant at most

$$\frac{\|X\|_{\mathrm{op}}}{\sqrt{nd}}. \tag{D.6.5}$$

Hence

$$\mathbb{P}(|g_j(b)| > u \mid X, Y, F_{-j}) \le 2\exp\left(-\frac{ndu^2}{2\|X\|_{\mathrm{op}}^2}\right). \tag{D.6.6}$$

This conditioning is valid for the compact restricted minimizer and for every deterministic restricted-GD time. Conditioning instead on a good event involving W_j before applying concentration would not be valid; we do not do so.

Choose a deterministic B_X with $\mathbb{P}(\|X\|_{\mathrm{op}} > B_X\sqrt{d}) \le e^{-c_X d}$ for large d; the same net proof permits $B_X = 8\sqrt{1+\alpha}$. For $d \ge 3$, define

$$T_d = 1 + \left\lceil \frac{\max\{0, \log(R_0 d)\}}{-\log q} \right\rceil, \qquad \tau_d = B_X\sqrt{\frac{8\log d}{n}}. \tag{D.6.7}$$

In particular $T_d = O(\log d)$ with a dimension-independent implied constant. Applying (D.6.6) and a union bound over $j \le m$, $0 \le t < T_d$, and the restricted minimizers gives

$$\max_{j, t<T_d} |g_j(b^{-j,t})| \le \tau_d, \qquad \max_j |g_j^*| \le \tau_d \tag{D.6.8}$$

except with probability at most

$$e^{-c_X d} + 2m(T_d + 1)d^{-4} = o(1). \tag{D.6.9}$$

No independence between different leave-one-feature problems or different times is asserted or required.

D.7 Uniform coefficient delocalization

First compare the full and restricted MLEs on the intersection of $\mathcal{G}_d$ and (D.6.8). Strong monotonicity along their segment gives

$$\mu\|\bar{b}^{-j} - b^*\|^2 \le \langle \nabla\mathcal{L}_n(\bar{b}^{-j}) - \nabla\mathcal{L}_n(b^*), \bar{b}^{-j} - b^*\rangle = -g_j^* b_j^*.$$

Cauchy–Schwarz implies

$$|b_j^*| \le \|\bar{b}^{-j} - b^*\| \le |g_j^*|/\mu, \quad \text{hence} \quad \|b^*\|_\infty \le \tau_d/\mu. \tag{D.7.1}$$

For the trajectories, let $G_d(b) = b - \eta_d\nabla\mathcal{L}_n(b)$. Equation (D.6.1) says

$$b^{-j,t+1} = G_d(b^{-j,t}) + \eta_d g_j(b^{-j,t})e_j.$$

The contraction established on the common ball gives

$$\|b^{t+1} - b^{-j,t+1}\| \le q_d\|b^t - b^{-j,t}\| + \eta_d|g_j(b^{-j,t})|. \tag{D.7.2}$$

Starting from zero and using $1 - q_d = \eta_d\mu$, (D.6.8) implies

$$\max_{j,\ 0\le t\le T_d} \|b^t - b^{-j,t}\| \le \frac{\eta_d\tau_d}{1-q_d} = \frac{\tau_d}{\mu}. \tag{D.7.3}$$

Since the restricted coordinate is zero, this bounds every $|b_j^t|$ through time T_d. For every $t \geq T_d$, (D.1.7) and the definition of T_d give

$$\|b^t\|_\infty \leq \|b^*\|_\infty + R_0 q^{T_d} \leq \tau_d/\mu + d^{-1}. \tag{D.7.4}$$

Thus on this single event,

$$\boxed{\sup_{t\geq 0} \|b^t\|_\infty \leq \tau_d/\mu + d^{-1}, \qquad \|b^*\|_\infty \leq \tau_d/\mu.} \tag{D.7.5}$$

Together with (D.4.4) and (D.6.9), this proves (D.1.8) with an explicit failure bound. The proof handles infinitely many GD times by deterministic contraction after a logarithmic horizon; it does not union-bound infinitely many concentration events.

D.8 Positive norms and simultaneous calibration floor

The lower-norm result is the argument in Appendix A Proposition 3; it is included here to make its use explicit. Put $K = HH^\top/m$, $\lambda_d = cd/n$, and $R_Y = Y - 1/2$. Then

$$\|b^1\|^2 = \frac{\lambda_d^2}{m} R_Y^\top K R_Y. \tag{D.8.1}$$

Conditional on X, F, decompose $R_Y = \bar{r} + \epsilon$, with independent centered bounded ϵ_i, and $D_{ii} = \mathbb{E}(\epsilon_i^2 \mid X, F) = \sigma'(\beta Z_i)$. A quadratic-form expansion gives

$$\mathrm{Var}(R_Y^\top K R_Y \mid X, F) \leq C\{\mathrm{tr}\, K^2 + \|K\bar{r}\|^2\} \leq Cn\|K\|_{\mathrm{op}}^2. \tag{D.8.2}$$

The norm $\|K\|$ is bounded in probability, so the normalization in (D.8.1) makes its centered fluctuation $o_P(1)$. The contribution $\bar{r}^\top K \bar{r}$ is nonnegative. Conditional on F, the bounded iid row variables $D_{ii}K_{ii}$ have expectation tending to $v\rho_\beta$: uniformly over features, $\|W_j\|^2 \to 1$ and $\theta^\top W_j \to 0$, and the relevant bounded bivariate Gaussian expectation is continuous. Hence

$$\frac{1}{n}\,\mathrm{tr}(KD) \to v\rho_\beta, \qquad \|b^1\|^2 \geq \frac{c^2 v\rho_\beta}{\alpha\gamma} + o_P(1). \tag{D.8.3}$$

Let $g_0 = \nabla \mathcal{L}_n(0)$. Descent and convexity respectively give, for every $t \geq 1$,

$$\mathcal{L}_n(b^t) \leq \log 2 - \frac{\eta_d}{2}\|g_0\|^2, \qquad \mathcal{L}_n(b^t) \geq \log 2 + g_0^\top b^t.$$

Therefore $\|b^t\| \geq \eta_d\|g_0\|/2 = \|b^1\|/2$, simultaneously at all positive times. Letting $t \to \infty$ gives the same bound for b^*. Equation (D.8.3) proves (D.1.9)–(D.1.10).

For a fresh test pair, condition on the complete training environment $\mathcal{E}_d = (F, X_{1:n}, Y_{1:n})$. Let $S_{t,d} = h(X)^\top b^t$ and

$$A_{t,d} = \mathbb{E}_X[ZS_{t,d} \mid \mathcal{E}_d] = g_F^\top b^t, \qquad Q_{t,d} = \mathbb{E}_X[S_{t,d}^2 \mid \mathcal{E}_d] = (b^t)^\top \Sigma_F b^t. \tag{D.8.4}$$

Use $t = \infty$ for b^*. Equations (D.2.1), (D.1.9), and Poincare yield, simultaneously for all $t \in \{1, 2, \ldots, \infty\}$,

$$Q_{t,d} - A_{t,d}^2 \geq sr_*^2 =: \Delta_* > 0, \qquad Q_{t,d} \leq L^2 R_0^2 =: Q_*. \tag{D.8.5}$$

The readout derivative bounds are uniform in time:

$$\|\nabla S_t\| \leq LR_0, \quad \|\nabla^2 S_t\|_{\mathrm{HS}} \leq 2L^2 R_0, \quad \|\nabla^2 S_t\|_{\mathrm{op}} \leq 2L^2 \sup_u \|b^u\|_\infty = o_P(1). \tag{D.8.6}$$

The proof of Appendix C Theorem 2.1 is sequentially uniform over classes with fixed bounds in (D.8.5)–(D.8.6). Otherwise a deterministic sequence violating uniformity would contradict that theorem. Consequently its random-environment argument gives

$$\sup_{t\in\{1,2,\dots,\infty\}} \left\| \mathcal{L}((Z, S_{t,d}) \mid \mathcal{E}_d) - N\left(0, \begin{pmatrix} 1 & A_{t,d} \\ A_{t,d} & Q_{t,d} \end{pmatrix}\right) \right\|_{\mathrm{TV}} \longrightarrow 0 \quad \text{in probability.} \tag{D.8.7}$$

This argument does not require deterministic overlap limits or a union bound over time.

For $Q > A^2$, define the explicit Gaussian calibration functional

$$\mathcal{C}_\beta(A, Q) = \mathbb{E}_U \left| \mathbb{E}_V \sigma\left(\beta \frac{A}{\sqrt{Q}} U + \beta \sqrt{1 - \frac{A^2}{Q}} V \right) - \sigma(\sqrt{Q} U) \right|, \qquad U, V \overset{\text{iid}}{\sim} N(0,1). \tag{D.8.8}$$

Let $\mathrm{CE}_{t,d}$ be exact, unbinned population calibration error of the predictor $\sigma(S_{t,d})$, conditional on $\mathcal{E}_d$. Since sigmoid is injective, conditioning on its probability is equivalent to conditioning on the score. the signed-measure argument in Appendix C makes this calibration error TV-Lipschitz. Applying the common label kernel $Y \mid Z$ and (D.8.7),

$$\sup_{t\in\{1,2,\dots,\infty\}} |\mathrm{CE}_{t,d} - \mathcal{C}_\beta(A_{t,d}, Q_{t,d})| \longrightarrow 0 \quad \text{in probability.} \tag{D.8.9}$$

For clarity, strict positivity of (D.8.8) holds for every $Q - A^2 > 0$. If $\beta = 0$, the true curve is the constant $1/2$ and the nonconstant prediction has positive error. If $\beta > 0, A \le 0$, the Gaussian conditional probability is nonincreasing while sigmoid is strictly increasing. If $\beta > 0, A > 0$, write its conditional curve as $g(s) = \mathbb{E}\sigma(ks + \tau V)$, with $k = \beta A/Q > 0$ and $\tau = \beta\sqrt{1 - A^2/Q} > 0$. Dominated convergence gives

$$\lim_{s\to\infty} e^{ks}(1 - g(s)) = e^{\tau^2/2}.$$

Equality with $\sigma(s)$ for every s would force first $k = 1$ and then $e^{\tau^2/2} = 1$, a contradiction. Equality almost everywhere would imply equality everywhere by continuity. Thus the error is strictly positive.

The parameter set

$$\mathcal{K} = \{(A, Q) : Q \le Q_*,\ Q - A^2 \ge \Delta_*\} \tag{D.8.10}$$

is compact. Formula (D.8.8) is continuous on it by dominated convergence. Therefore

$$\varepsilon_* := \min_{(A,Q)\in\mathcal{K}} \mathcal{C}_\beta(A, Q) > 0. \tag{D.8.11}$$

Equations (D.8.5) and (D.8.9) prove the simultaneous statement

$$\mathbb{P}\left(\inf_{t\in\{1,2,\dots,\infty\}} \mathrm{CE}_{t,d} \ge \varepsilon_*/2 \right) \longrightarrow 1. \tag{D.8.12}$$

In particular, **no training-measurable stopping rule taking a positive integer time, even one diverging with dimension, attains asymptotic calibration in this fixed-step regime.** The same holds if the rule may output the MLE. At time zero the constant predictor $1/2$ is calibrated, since the symmetric teacher has marginal class probability $1/2$. The theorem does not cover steps $c = c_d \to 0$, which alter the stated regime.

Positive absolute calibration error does not alone assert that every confidence level is overconfident. The full curve in (D.8.8), or its conditional form $g(s)$, determines the direction. Early positive iterates can be underconfident while still having strictly positive exact calibration error.

D.9 Promotion of deterministic finite-time overlaps to all times

Theorem A and Appendix B prove, for every fixed integer $t \geq 1$, deterministic limits

$$A_{t,d} \xrightarrow{P} A_t(\alpha, \gamma, c, \beta), \qquad Q_{t,d} \xrightarrow{P} Q_t(\alpha, \gamma, c, \beta). \tag{D.9.1}$$

Then these limits extend to $t = \infty$, and convergence is uniform over all positive integer times and the MLE:

$$\sup_{t \in \{1,2,\ldots,\infty\}} \{|A_{t,d} - A_t| + |Q_{t,d} - Q_t|\} \xrightarrow{P} 0. \tag{D.9.2}$$

Here is the proof, including the exchange of limits. The population norm bounds are $\|g_F\| \leq L$ and $\|\Sigma_F\| \leq L^2$. From (D.1.7), for every $t \geq T$, including infinity,

$$\|b^t - b^T\| \leq 2R_0 q^T,$$

$$|A_{t,d} - A_{T,d}| \leq 2LR_0 q^T, \qquad |Q_{t,d} - Q_{T,d}| \leq 4L^2 R_0^2 q^T. \tag{D.9.3}$$

For each fixed t, T, take probability limits in these deterministic inequalities on events of probability tending to one. This gives the same bound between (A_t, Q_t) and (A_T, Q_T). Therefore the deterministic sequence is exponentially Cauchy and has a limit (A_∞, Q_∞). For arbitrary fixed T, split the supremum in (D.9.2) into the finite times below T and the common exponential tail. Equation (D.9.1) controls the finite part, while (D.9.3) controls both tails. Let $d \to \infty$, then $T \to \infty$. This proves (D.9.2) without interchanging unproved limits.

Combining uniform continuity on (D.8.10), (D.8.9), and (D.9.2) gives the deterministic whole-trajectory theorem

$$\boxed{\sup_{t \in \{1,2,\ldots,\infty\}} \left|\mathrm{CE}_{t,d} - \mathcal{C}_\beta\big(A_t(\alpha, \gamma, c, \beta), Q_t(\alpha, \gamma, c, \beta)\big)\right| \xrightarrow{P} 0.} \tag{D.9.4}$$

For a data-dependent positive stopping rule τ_d, the same uniform estimate holds after substituting $t = \tau_d$. If $\tau_d \to \infty$ in probability, then $\mathrm{CE}_{\tau_d,d} \to \mathcal{C}_\beta(A_\infty, Q_\infty) > 0$ in probability. If the stopping time has no limiting value, the deterministic curve still approximates its realized index and the common positive floor still applies.

The finite-time result (D.9.1) is what makes (D.9.4) a calibration function of the aspect ratios and training time. Equations (D.8.7)–(D.8.12) themselves do not require that input and are already unconditional consequences of the proofs above and the analytic transfer theorem in Appendix C.

Appendix E. A single-crossing theorem and signed overconfidence

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

E.1 Statement

Let Z, G be independent standard normal variables, let $k, \tau, \beta > 0$, and put

$$U = kZ + \tau G, \qquad V = k^2 + \tau^2. \tag{E.1}$$

Given Z, let Y be Bernoulli with mean $\sigma(\beta Z)$, independently of G, where $\sigma(x) = (1 + e^{-x})^{-1}$. Its conditional regression given the score coordinate is

$$g(u) := \mathbb{E}[Y \mid U = u] = \mathbb{E}_H \sigma(\lambda u + sH), \qquad \lambda = \frac{\beta k}{V}, \quad s = \frac{\beta\tau}{\sqrt{V}}, \tag{E.2}$$

with H standard normal. Consider the positive-scale predictors $p_c(u) = \sigma(cu)$ and their population cross-entropy

$$R(c) = \mathbb{E}\{\log(1 + e^{cU}) - cUg(U)\}. \tag{E.3}$$

Theorem. For every $k, \tau, \beta > 0$:

1. The function
$$h(u) = \frac{\operatorname{logit} g(u)}{u}, \qquad u > 0, \tag{E.4}$$
is continuous and strictly increasing, with endpoint limits
$$h(0+) = 4g'(0) = 4\lambda\mathbb{E}\sigma'(sH), \qquad h(+\infty) = \lambda. \tag{E.5}$$

2. R has a unique global minimizer c_*, and
$$4g'(0) < c_* < \lambda, \qquad R(c_*) < R(0) = \log 2. \tag{E.6}$$

3. There is a unique $u_* > 0$ such that
$$\sigma(c_* u) - g(u) \begin{cases} > 0, & 0 < u < u_*, \\ = 0, & u = u_*, \\ < 0, & u > u_*. \end{cases} \tag{E.7}$$

4. The average signed overconfidence is strictly positive:
$$\mathrm{OC}(c_*) := \mathbb{E}\big[(\sigma(c_* U) - g(U)) \operatorname{sign} U\big] > 0. \tag{E.8}$$

Here the predicted class is $\mathbf{1}_{\{U>0\}}$, its confidence is $\sigma(c_*|U|)$, and its conditional probability of being correct is $g(|U|)$. With $S = c_* U$, conditioning on U gives the equivalent exact definition

$$\mathrm{OC}(c_*) = \mathbb{E}[(\sigma(S) - \sigma(\beta Z)) \operatorname{sign} S].$$

Thus (E.8) is exactly mean confidence minus mean accuracy. It is not a claim of overconfidence at every score: (E.7) proves underconfidence in the far tails.

E.2 Logistic noise as a normal variance mixture

Let $E_1, E_2, \ldots$ be independent exponential variables of mean one, and define

$$W = 2\sum_{j=1}^{\infty} \frac{E_j}{j^2}. \tag{E.9}$$

The series is finite almost surely, because its expectation is $2\sum j^{-2} = \pi^2/3 < \infty$. If N is an independent standard normal variable, the characteristic function of $\sqrt{W}N$ is

$$\mathbb{E}e^{-t^2W/2} = \prod_{j=1}^{\infty}(1 + t^2/j^2)^{-1} = \frac{\pi t}{\sinh(\pi t)}, \tag{E.10}$$

with the continuous value one at $t = 0$. The last identity is Euler's product for sinh. On the other hand, the characteristic function of a standard logistic variable L is

$$\int_{-\infty}^{\infty} e^{itx}\frac{e^x}{(1+e^x)^2}\,dx = \int_0^{\infty}\frac{r^{it}}{(1+r)^2}\,dr = \Gamma(1+it)\Gamma(1-it) = \frac{\pi t}{\sinh(\pi t)}. \tag{E.11}$$

The middle equality is the beta integral, and the last follows from the gamma reflection identity. Uniqueness of characteristic functions therefore proves

$$L \overset{d}{=} \sqrt{W}N. \tag{E.12}$$

We also need strict support, not just a formal mixture. Write $W = 2E_1 + Q$, where $Q = 2\sum_{j\geq 2} E_j/j^2$. For every $w > 0$, $\mathbb{P}(Q < w) > 0$: choose a finite cutoff whose remaining tail has expectation less than $w/4$; the tail is below $w/2$ with positive probability by Markov's inequality, and the finite initial sum is below $w/2$ with positive probability, independently. Conditioning on Q gives the density

$$f_W(w) = \frac{1}{2}e^{-w/2}\mathbb{E}\big[e^{Q/2}\mathbf{1}_{\{Q<w\}}\big] > 0 \quad (w > 0). \tag{E.13}$$

It is finite, since the integrand is bounded by $e^{w/2}$. Hence the CDF F_W is strictly increasing on $(0, \infty)$, is positive at every positive argument, and vanishes at nonpositive arguments.

E.3 A direct strict single-crossing argument

For $s > 0$, write

$$F_s(x) = \mathbb{E}\sigma(x + sH), \qquad \Lambda_s(x) = \operatorname{logit} F_s(x). \tag{E.14}$$

By symmetry of H, F_s is the CDF of $L + sH$. Conditional on W, this is a centered normal variable of variance $W + s^2$. Consequently

$$F_s(x) = \mathbb{E}\Phi\left(\frac{x}{\sqrt{W+s^2}}\right), \qquad \sigma(cx) = \mathbb{E}\Phi\left(\frac{x}{\sqrt{W}/c}\right) \quad (c > 0). \tag{E.15}$$

Set $x = e^t$. The difference of the CDFs of the logarithms of the two random scales in (E.15) is

$$N_c(u) = F_W(e^{2u} - s^2) - F_W(c^2 e^{2u}). \tag{E.16}$$

Integration by parts yields

$$D_c(t) := F_s(e^t) - \sigma(ce^t) = \int_{\mathbb{R}} K(t-u) N_c(u)\, du, \qquad K(v) = e^v \varphi(e^v). \tag{E.17}$$

For completeness, the function being integrated against the signed scale law is $u \mapsto \Phi(e^{t-u})$; its derivative is $-K(t-u)$. The boundary terms vanish because $N_c(u) \to 0$ at both ends. The integral is absolutely convergent: $|N_c| \le 1$ and $\int K(v)\, dv = 1/2$.

If $c \ge 1$, (E.13) and (E.16) give $N_c(u) < 0$ for every finite u; hence $D_c(t) < 0$ for every t. In particular

$$\tfrac{1}{2} < F_s(x) < \sigma(x) \quad (x > 0). \tag{E.18}$$

The first strict inequality follows from symmetry and strict monotonicity of F_s.

For $0 < c < 1$, put

$$u_c = \frac{1}{2} \log \frac{s^2}{1 - c^2}. \tag{E.19}$$

Strict monotonicity of F_W implies

$$N_c(u) < 0 \ (u < u_c), \qquad N_c(u_c) = 0, \qquad N_c(u) > 0 \ (u > u_c). \tag{E.20}$$

For any $t_2 > t_1$, the kernel ratio

$$Q(u) := \frac{K(t_2 - u)}{K(t_1 - u)} = e^{t_2 - t_1} \exp\left[-\frac{e^{2t_2} - e^{2t_1}}{2} e^{-2u} \right] \tag{E.21}$$

is strictly increasing in u, and is bounded above by $e^{t_2 - t_1}$. Therefore

$$D_c(t_2) - Q(u_c) D_c(t_1) = \int_{\mathbb{R}} K(t_1 - u)[Q(u) - Q(u_c)] N_c(u)\, du > 0. \tag{E.22}$$

Every factor except the bracket and N_c is positive; these last two factors have the same strict sign on either side of u_c. The integral is finite by the bound on Q and the integrability of K. In particular, a zero of D_c is followed by strictly positive values. This proves the exact direction of the required single crossing without an appeal to a general variation-diminishing result.

Now fix $0 < x_1 < x_2$, and choose

$$c = \frac{\Lambda_s(x_1)}{x_1} \in (0, 1), \tag{E.23}$$

where membership in $(0, 1)$ follows from (E.18). Then $D_c(\log x_1) = 0$. Equation (E.22) gives

$$F_s(x_2) > \sigma(cx_2), \quad \text{so} \quad \frac{\Lambda_s(x_2)}{x_2} > \frac{\Lambda_s(x_1)}{x_1}. \tag{E.24}$$

This is strict monotonicity for every pair $x_1 < x_2$, without needing an analyticity argument.

At zero, dominated differentiation and $F_s(0) = 1/2$ give

$$\lim_{x \downarrow 0} \frac{\Lambda_s(x)}{x} = 4\mathbb{E}\sigma'(sH) \in (0, 1). \tag{E.25}$$

For the other endpoint,

$$e^x[1 - F_s(x)] = \mathbb{E}\frac{e^{-sH}}{1 + e^{-x-sH}} \longrightarrow e^{s^2/2} \tag{E.26}$$

by dominated convergence, since e^{-sH} is integrable. Thus

$$\Lambda_s(x) = x - \frac{s^2}{2} + o(1), \qquad \lim_{x \to \infty} \frac{\Lambda_s(x)}{x} = 1. \tag{E.27}$$

Applying (E.24)–(E.27) with $x = \lambda u$ proves (E.4)–(E.5).

E.4 Risk minimization and strict signed overconfidence

The normal law of U has all moments, so dominated differentiation gives

$$R'(c) = \mathbb{E}[U(\sigma(cU) - g(U))], \qquad R''(c) = \mathbb{E}[U^2\sigma'(cU)] > 0. \tag{E.28}$$

Since $g(u) > 1/2$ for $u > 0$, symmetry implies $R'(0) < 0$. On the other hand,

$$\lim_{c \to \infty} R'(c) = \mathbb{E}[|U|(1 - g(|U|))] > 0. \tag{E.29}$$

The integrand is strictly positive almost surely and is integrable. Thus the strictly convex function R has a unique critical point $c_* > 0$, and it is its global minimizer. Its value is strictly below $R(0) = \log 2$.

Let $h_0 = 4g'(0)$. By the strict increase already proved, $h_0 < h(u) < \lambda$ for every $u > 0$. Hence $R'(c) < 0$ for $c \le h_0$, and $R'(c) > 0$ for $c \ge \lambda$. This proves $h_0 < c_* < \lambda$. Continuity, strict increase and the endpoint limits produce a unique $u_* > 0$ with $h(u_*) = c_*$, and (E.7) follows from the strict increase of σ.

Write $e(u) = \sigma(c_*u) - g(u)$, and let φ_V denote the $N(0, V)$ density. The first-order condition and oddness of e give

$$0 = R'(c_*) = 2\int_0^\infty u\, e(u)\varphi_V(u)\, du. \tag{E.30}$$

Subtracting this identity divided by u_* from the unweighted signed error gives

$$\begin{aligned}\mathrm{OC}(c_*) &= 2\int_0^\infty e(u)\varphi_V(u)\,du \\ &= 2\int_0^\infty e(u)\left(1-\frac{u}{u_*}\right)\varphi_V(u)\,du > 0.\end{aligned} \tag{E.31}$$

The two factors $e(u)$ and $1-u/u_*$ have the same strict sign for every $u>0$ other than u_*. Integrability follows from $|e|\le 1$ and the first absolute normal moment. This completes the theorem.

One may compute c_* through a one-dimensional strictly monotone equation. Stein's identity and the original representation (E.1) give

$$V c_* \,\mathbb{E}\sigma'(c_*\sqrt{V}\,H) = k\beta\,\mathbb{E}\sigma'(\beta H). \tag{E.32}$$

Equation (E.28), rather than any presumed monotonicity of the individual factors in (E.32), proves uniqueness.

E.5 Consequence for the random-feature model

Suppose a first-step Gaussian random-feature theorem gives, conditional on the trained model,

$$\mathcal{L}(Z, U_d \mid \text{model}) \longrightarrow \mathcal{L}(Z, kZ+\tau G) \quad \text{in total variation in probability,} \tag{E.33}$$

with $k,\tau>0$, and gives population cross-entropy convergence at every fixed deterministic scale. Taking the fixed scale $c=c_*$ from this appendix then yields

$$\mathrm{OC}_d(c_*) \xrightarrow{\mathbb{P}} \mathrm{OC}(c_*) > 0, \qquad R_d(c_*) \xrightarrow{\mathbb{P}} R(c_*) < \log 2. \tag{E.34}$$

Indeed signed OC is the conditional expectation of the bounded measurable function

$$(z,u) \longmapsto [\sigma(c_*u)-\sigma(\beta z)]\,\mathrm{sign}\,u, \tag{E.35}$$

so total variation proves the first assertion directly. A second moment bound or the separately established risk-transfer theorem handles the unbounded cross-entropy integrand. In particular, fixed positive margins below $\log 2$ and above zero hold simultaneously with probability tending to one. Theorem A and Appendix C supply these total-variation and risk limits.

Because $u\mapsto\sigma(c_*u)$ is one-to-one, the limiting exact unbinned calibration error is

$$\mathbb{E}|\sigma(c_*U)-g(U)| \ge \mathrm{OC}(c_*) > 0. \tag{E.36}$$

The finite-model OC statement (E.34) itself does not require continuity of conditional regression: it uses the bounded test function (E.35). This is weaker in prerequisite than proving convergence of an entire conditional calibration curve.

The theorem treats all positive parameters in the Gaussian scalar experiment, not a selected categorical surrogate or one numerical parameter choice. It closes a signed-overconfidence proof obligation at a risk-optimal **first-step scale**. It does not establish a deterministic arbitrary-time neural-GD trajectory, interchange a training-time limit with dimension, optimize over all stopping times, or show that the scalar risk optimum is a global optimum over the full

GD path. Those remain separate the present problem obligations. The limiting predictor is overconfident near the boundary and underconfident in the tails, so a claim of pointwise overconfidence at all scores would be false.

E.6 An explicit common regime with the whole-trajectory stopping theorem

The risk-optimal scale above is not automatically a stable GD step. Nevertheless, there is an explicit nonempty regime in which it satisfies the small-step and sample-size conditions of the entire-trajectory theorem in Appendix D. Thus the overconfidence and stopping conclusions need not be about incompatible parameter choices.

Use the actual first-step law from the Gaussian tanh RF calculation, with the step scale factored out:

$$S_1 = cU, \qquad k = a^2\mu_\beta, \qquad \tau^2 = \frac{\mu_\beta^2 a^2 v}{\gamma} + \frac{a^4 + v^2/\gamma}{4\alpha}, \tag{E.37}$$

where

$$a = \mathbb{E}[H \tanh H] > 0, \quad v = \mathbb{E}\tanh^2 H > 0, \quad \nu = v - a^2 > 0,$$
$$\mu_\beta = \mathbb{E}[H(\sigma(\beta H) - 1/2)] = \beta \mathbb{E}\sigma'(\beta H).$$

These are exactly the coefficients in Appendix B, equation (B.21), and in equation (11) of the main text. For $\beta > 0$, both k and τ are strictly positive.

Fix any $\gamma > 0$, and use the constants of Appendix D

$$L = 8\sqrt{1+\gamma}, \qquad s_0 = \nu/2, \qquad \kappa = \sqrt{s_0/2}, \qquad p_0 = \frac{s_0^2}{20L^4}, \qquad B = \sqrt{2\log(4/p_0)}.$$

Choose

$$\frac{\alpha}{\gamma} > C_0 := \max\left\{1, \left(\frac{256L}{\kappa p_0}\right)^2\right\}, \qquad 0 < \beta < \min\left\{\frac{\log 3}{B}, \frac{v}{8aL\sqrt{\alpha}}\right\}. \tag{E.38}$$

Every interval here has positive length. In particular this is a nonempty open set of proportional sample, width and teacher parameters, although its sample ratio is deliberately very conservative.

The first inequality on β gives $q_\beta = \sigma(-\beta B) > 1/4$ and hence $p_1 = q_\beta p_0/2 > p_0/8$ in Appendix D. Thus its threshold satisfies

$$C_*(\beta) = \max\left\{1, \left(\frac{32L}{\kappa p_1}\right)^2\right\} \le C_0 < \alpha/\gamma. \tag{E.39}$$

Next, $\mu_\beta \le \beta/4$ and (E.37) give

$$0 < c_* < \frac{\beta k}{k^2 + \tau^2} \le \frac{a^2\alpha\gamma\beta^2}{v^2} < \frac{\gamma}{64L^2} < \frac{\gamma}{32L^2} = \frac{\gamma}{2\Lambda}, \qquad \Lambda = 16L^2. \tag{E.40}$$

This is precisely inside the stable fixed-step interval of Appendix D. The scalar risk optimum is therefore a valid fixed deterministic GD parameter in the same model for which Appendix D proves its all-time statements.

Consequently, Theorems A and B imply that this one model simultaneously has:

- A first positive iterate with population risk strictly below $\log 2$ and strictly positive signed overconfidence, with probability tending to one.
- A fixed positive exact-calibration floor, with probability tending to one, simultaneously over every positive integer training time, the MLE, and any training-measurable stopping rule taking those values.

- A calibrated time-zero predictor with risk exactly $\log 2$.

This also gives a nonvacuous risk-constrained stopping conclusion. For any fixed $\Delta > 0$ and any training-measurable rule $T_d \in \{0, 1, \ldots, \infty\}$ such that

$$\mathbb{P}\{R_d(T_d) \leq \log 2 - \Delta\} \longrightarrow 1, \tag{E.41}$$

one necessarily has $\mathbb{P}(T_d > 0) \to 1$, because the risk at time zero is exactly $\log 2$. The simultaneous positive calibration floor then applies to T_d. On the other hand, a fixed first-step rule satisfies (E.41) for every $\Delta < R(0) - R(c_*)$ by (E.34). Thus such risk improvement is attainable but cannot coexist with asymptotic exact calibration through stopping alone in this fixed-step regime.

This common regime still has sufficiently large n/m, so the fitted output layer is underparameterized. The random-feature architecture remains the original dense Gaussian nonlinear model; no categorical design is introduced. The argument does not extend the all-time statement to every overparameterized sample ratio or to dimension-dependent vanishing steps.

Appendix F. All-time obstruction in the strongly overparameterized regime

Theorem, lemma, and proposition numbers in this appendix are local; equation and section references carry the appendix letter.

F.1 Model and explicit nonempty regime

Use the exact original Gaussian tanh model:

$$X_i \sim N(0, I_d), \quad W_j \sim N(0, I_d/d), \quad Y_i \mid X_i \sim \text{Bernoulli}(\sigma(\beta\theta^\top X_i)),$$

with independent inputs and feature rows, conditionally independent labels, $\|\theta\| = 1$, and fixed finite $\beta \geq 0$. Let

$$n/d \to \alpha > 0, \qquad m/d \to \gamma > 0, \quad h(x) = \tanh(Fx), \quad H_{ij} = h_j(X_i).$$

Only the output coefficients are trained. Put $s_i = 2Y_i - 1$, $\ell(z) = \log(1 + e^{-z})$, and

$$\mathcal{L}_n(b) = \frac{1}{n}\sum_i \ell(s_i h_i^\top b), \qquad b^0 = 0, \quad b^{t+1} = b^t - \eta_d \nabla \mathcal{L}_n(b^t), \quad \eta_d = \frac{cd}{m}. \tag{F.1.1}$$

Thus the actual update coefficient multiplying $H^\top[Y - \sigma(Hb^t)]$ is $cd/(nm)$, as in the main theorem. The scale $c > 0$ is fixed independently of dimension.

Define

$$a = \mathbb{E}[G \tanh G], \quad v = \mathbb{E}\tanh^2 G, \quad \nu = v - a^2 > 0, \quad s = \nu/2, \quad \kappa = \sqrt{s/2}, \quad G \sim N(0,1).$$

For the training-input side, set

$$L_X = 8\sqrt{1+\alpha}, \quad p_X = \frac{s^2}{20L_X^4}, \quad C_X = \max\left\{1, \left(\frac{32L_X}{\kappa p_X}\right)^2\right\},$$

$$k_0 = \frac{\kappa^2 p_X}{16}, \qquad k_1 = 64L_X^2. \tag{F.1.2}$$

The explicit assumptions for this module are

$$\boxed{\gamma/\alpha > C_X, \qquad 0 < c < 2\alpha/k_1.} \tag{F.1.3}$$

For each fixed $\alpha > 0$ and finite β, these conditions leave nonempty open intervals of width ratios and learning scales. They imply $m > n$ for large dimension. Their conservative constants are not claimed to be sharp.

For the fresh-input feature side, define

$$L_F = 8\sqrt{1+\gamma}, \quad p_F = \frac{s^2}{20L_F^4}, \quad B_F = \sqrt{2\log(4/p_F)}, \quad q_F = \sigma(-\beta B_F) > 0,$$

$$C_F = \log\frac{4}{q_F p_F}, \quad R_F = \frac{C_F}{\kappa}, \quad \varepsilon_F = \frac{q_F p_F}{4}, \quad a_F = \frac{q_F \kappa p_F}{2} > 0. \tag{F.1.4}$$

Finally put

$$D = 2\sqrt{\gamma k_1/\alpha}, \qquad \mathfrak{B}_t = \frac{1}{t+1} + \frac{\alpha}{ck_0}\frac{\log^2(t+1)}{t} \quad (t \geq 1). \tag{F.1.5}$$

These are all deterministic dimension-independent constants or sequences.

F.2 The simultaneous design event, proved for the actual nonlinear features

With probability tending to one, all of the following hold:

$$\|F\|_{\mathrm{op}} \le L_F, \qquad \Sigma_F := \mathbb{E}_X[h(X)h(X)^\top] \succeq sI_m, \tag{F.2.1}$$

$$k_0 I_n \preceq HH^\top/m \preceq k_1 I_n. \tag{F.2.2}$$

We prove these facts directly for the actual nonlinear features.

F.2.1 Population covariance on either side

Consider a matrix V with r independent rows $N(0, I_d/d)$, where r/d tends to a fixed positive constant. For an independent $G \sim N(0, I_d)$, write $h_V(G) = \tanh(VG)$. If $q_j = \|V_j\|^2$ and $A(q) = \mathbb{E}\tanh'(\sqrt{q}G_1)$, the Hermite expansion gives

$$\mathbb{E}[h_V h_V^\top] = D_A VV^\top D_A + D_\nu + E, \tag{F.2.3}$$

where

$$(D_\nu)_{jj} = \mathbb{E}\tanh^2(\sqrt{q_j}G_1) - q_j A(q_j)^2, \quad E_{jj} = 0, \quad |E_{ij}| \le |\rho_{ij}|^3 \ (i \ne j).$$

Here ρ_{ij} is the inner product of the two normalized rows. The linear Hermite coefficient is identified by Gaussian integration by parts; all remaining degrees are odd and at least three. Cauchy–Schwarz on their coefficients gives the displayed remainder bound, since the second moment of tanh is at most one. Independent Gaussian row directions satisfy

$$\mathbb{E}\|E\|_F^2 \le \frac{15r(r-1)}{d(d+2)(d+4)} = O(d^{-1}). \tag{F.2.4}$$

Uniform row-norm concentration makes $D_\nu \to \nu I$ in operator norm, while (F.2.4) makes $\|E\|_{\mathrm{op}} \to 0$ in probability. Thus the covariance is at least sI with probability tending to one. Gaussian sphere nets also give $\|V\|_{\mathrm{op}} \le 8\sqrt{1 + r/d + o(1)}$ with exponential probability; the stated constants L_F, L_X have ample fixed slack. More explicitly, the covariance event can be chosen with failure $O(d^{-1}) + O(de^{-c_0 d})$.

Apply this first to $V = F$, obtaining (F.2.1). Apply it also to $V = X/\sqrt{d}$, obtaining an event $\mathcal{X}_d$, measurable in the training inputs alone, on which

$$\|X\|_{\mathrm{op}}/\sqrt{d} \le L_X, \qquad K_X := \mathbb{E}_G[\tanh(XG/\sqrt{d})\tanh(XG/\sqrt{d})^\top] \succeq sI_n. \tag{F.2.5}$$

This second use is legitimate because the rows $X_i/\sqrt{d}$ are iid Gaussian. It does not replace the actual feature matrix by independent entries.

F.2.2 Conditional independent columns and small balls

Condition on $X \in \mathcal{X}_d$. The m columns h_j of H are iid centered vectors of the form $\tanh(XG_j/\sqrt{d})$. For every unit $u \in \mathbb{R}^n$, $u^\top h_j$ is a centered L_X-Lipschitz function of a standard Gaussian vector. Consequently

$$\mathbb{E}e^{zu^\top h_j} \le e^{z^2 L_X^2/2}, \quad \mathbb{E}(u^\top h_j)^2 \ge s, \quad \mathbb{E}(u^\top h_j)^4 \le 5L_X^4. \tag{F.2.6}$$

The fourth-moment bound follows by Gaussian Poincare applied to the function and its square. Paley–Zygmund then gives

$$\mathbb{P}(|u^\top h_j| \ge \kappa \mid X) \ge p_X. \tag{F.2.7}$$

Let $\psi(z) = 0$ for $|z| \le \kappa/2$, let it rise linearly to one on $\kappa/2 < |z| < \kappa$, and let it equal one for $|z| \ge \kappa$. It is $2/\kappa$-Lipschitz, bounded by one, and vanishes at zero. Symmetrization, Rademacher contraction, and $\operatorname{tr} K_X \le nL_X^2$ give

$$\mathbb{E} \sup_{\|u\|=1} \left| \frac{1}{m} \sum_j \psi(u^\top h_j) - \mathbb{E}\psi(u^\top h_j) \right| \le \frac{8L_X}{\kappa} \sqrt{\frac{n}{m}}. \tag{F.2.8}$$

If $m/n \ge C_X$, this is at most $p_X/4$. Bounded differences with deviation $p_X/4$ proves, except with conditional probability $e^{-mp_X^2/8}$,

$$\inf_{\|u\|=1} \frac{1}{m} \#\{j : |u^\top h_j| > \kappa/2\} \ge p_X/2. \tag{F.2.9}$$

In fact (F.2.9) is robust: after deleting any subset of at most $p_X m/4$ columns, the remaining Gram matrix, still normalized by m, is at least $\kappa^2 p_X I/16 = k_0 I$. No independence of the deleted columns or union bound over subsets is needed. In particular it proves the lower half of (F.2.2).

The same two-sphere-net bound applied to $H^\top$ gives

$$\|H\|_{\mathrm{op}} \le 8L_X \sqrt{m} \tag{F.2.10}$$

with conditional failure at most $2e^{-3m}$ when $m \ge n$: each fixed bilinear form is centered L_X-subGaussian, and $1/4$-nets have cardinalities at most $9^m, 9^n$. This proves the upper half of (F.2.2).

Let $\mathcal{G}_d$ combine (F.2.1), (F.2.5), and (F.2.2). For large d, (F.1.3) ensures $m/n \ge C_X$, and

$$\mathbb{P}(\mathcal{G}_d^c) \le O(d^{-1}) + O(de^{-c_0 d}) + e^{-mp_X^2/8} + 2e^{-3m} = o(1). \tag{F.2.11}$$

No independence between the good feature event and the conditional sample-Gram event is asserted: a union bound suffices. Labels play no role in this design event.

F.3 Uniform empirical loss decay

On $\mathcal{G}_d$, the full row rank in (F.2.2) makes every binary labeling strictly separable. For $R > 0$, the explicit comparator

$$b_R = H^\top (HH^\top)^{-1} (Rs_{1:n}) \tag{F.3.1}$$

has every signed training margin equal to R, and

$$\mathcal{L}_n(b_R) = \ell(R) \le e^{-R}, \qquad \|b_R\|^2 \le \frac{nR^2}{mk_0}. \tag{F.3.2}$$

The empirical Hessian is globally bounded by

$$\nabla^2 \mathcal{L}_n(b) \preceq \frac{H^\top H}{4n}, \quad \text{so its smoothness constant is at most } \frac{mk_1}{4n}.$$

For large d, $n/d \ge \alpha/2$; assumption (F.1.3) gives

$$\eta_d \frac{mk_1}{4n} = \frac{cdk_1}{4n} < 1. \tag{F.3.3}$$

Hence ordinary GD decreases the empirical loss at every step.

For clarity, the convex smooth-GD comparator inequality used here is

$$\mathcal{L}_n(b^t) - \mathcal{L}_n(u) \le \frac{\|u\|^2}{2\eta_d t} \quad (t \ge 1). \tag{F.3.4}$$

It follows directly by expanding $\|b^{k+1} - u\|^2$, using convexity, and using descent to replace $\eta_d^2 \|\nabla \mathcal{L}_n(b^k)\|^2$ by $2\eta_d[\mathcal{L}_n(b^k) - \mathcal{L}_n(b^{k+1})]$. Summing gives the bound on the average of losses at times $1, \dots, t$, and monotonicity bounds the final loss by that average. No strong convexity or finite MLE is needed.

Use (F.3.2) in (F.3.4), then take $R = \log(t+1)$. Since $n/d \le 2\alpha$ eventually,

$$\boxed{\mathcal{L}_n(b^t) \le \frac{1}{t+1} + \frac{n}{2cdk_0} \frac{\log^2(t+1)}{t} \le \mathfrak{B}_t \qquad \text{for every } t \ge 1.} \tag{F.3.5}$$

This inequality holds simultaneously at all integer times on one event. In particular it applies to times growing arbitrarily quickly with dimension.

F.4 Uniform norm growth and a deterministic crossing time

For any b, Jensen's inequality, monotonicity of ℓ, and Cauchy–Schwarz give

$$\mathcal{L}_n(b) \ge \ell\left(\frac{1}{n}\sum_i s_i h_i^\top b\right) \ge \ell\left(\frac{\|Hb\|}{\sqrt{n}}\right) \ge \ell(D\|b\|). \tag{F.4.1}$$

The last inequality uses $\|H\|^2/m \le k_1$ and $m/n \le 4\gamma/\alpha$ for large d. Thus (F.3.5) implies

$$\boxed{\|b^t\| \ge \frac{1}{D}\left[-\log(e^{\mathfrak{B}_t} - 1)\right]_+ \quad (t \ge 1).} \tag{F.4.2}$$

Its right-hand side grows as $D^{-1}(\log t - 2\log\log t - O(1))$. The path need not have monotone norm; the simultaneous lower bound is what is used.

For any deterministic threshold $R_* > 0$, define the explicit finite integer

$$T(R_*) = \min\{T \ge 2 : \mathfrak{B}_T < \ell(DR_*)\}. \tag{F.4.3}$$

It exists because $\mathfrak{B}_T \to 0$. At this time the loss is below $\ell(DR_*)$; monotone empirical loss and (F.4.1) then give

$$\|b^t\| > R_* \qquad \text{for every integer } t \ge T(R_*). \tag{F.4.4}$$

The time $T(R_*)$ depends only on the fixed model parameters and the chosen threshold, never on dimension or the data. This is stronger than taking $t \to \infty$ separately for each fixed dataset.

F.5 A calibration floor for every sufficiently large readout

We prove a large-norm calibration witness directly. On (F.2.1), every coefficient vector b, including arbitrary training-dependent vectors, satisfies

$$\mathbb{E}_X S_b^2 \ge s\|b\|^2, \qquad \mathbb{E}_X S_b^4 \le 5L_F^4\|b\|^4, \quad S_b = b^\top h(X).$$

Consequently

$$\mathbb{P}_X(|S_b| \ge \kappa\|b\|) \ge p_F \quad \text{for every } b \ne 0. \tag{F.5.1}$$

All constants and inequalities are simultaneous over b. This statement follows from uniform population moment bounds; it does not union-bound over coefficient vectors.

Write $Z = \theta^\top X$, $p(Z) = \sigma(\beta Z)$. The choice of B_F ensures $\mathbb{P}(|Z| > B_F) \le p_F/2$, and on the complementary event either label has probability at least q_F. Define the bounded measurable score witness

$$f(S) = \operatorname{sign}(S)\mathbf{1}_{\{|S|\ge C_F\}}.$$

For every score and teacher coordinate,

$$[\sigma(S) - p(Z)]f(S) \ge q_F\mathbf{1}_{\{|S|\ge C_F,\ |Z|\le B_F\}} - e^{-C_F}\mathbf{1}_{\{|S|\ge C_F\}}. \tag{F.5.2}$$

Indeed, on a positive score the signed difference is at least $1 - p(Z) - e^{-C_F}$, and on a negative score it is at least $p(Z) - e^{-C_F}$. Outside the bounded-teacher event the remaining lower bound is still $-e^{-C_F}$.

If $\|b\| \ge R_F = C_F/\kappa$, (F.5.1) implies $\mathbb{P}(|S_b| \ge C_F, |Z| \le B_F) \ge p_F/2$. Hence

$$\mathbb{E}_X[(\sigma(S_b) - p(Z))f(S_b)] \ge q_F p_F/2 - e^{-C_F} \ge \varepsilon_F. \tag{F.5.3}$$

By the defining signed-measure characterization of exact calibration error, or just conditional expectation and $|f| \le 1$,

$$\mathrm{CE}(b) = \mathbb{E}_X\left|\mathbb{E}[p(Z) \mid S_b] - \sigma(S_b)\right| \ge \left|\mathbb{E}_X[(\sigma(S_b) - p(Z))f(S_b)]\right|.$$

Therefore

$$\boxed{\inf_{\|b\|\ge R_F} \mathrm{CE}(b) \ge \varepsilon_F > 0.} \tag{F.5.4}$$

This holds on the good feature event for all readouts at once. It does not require their coefficients to be delocalized or their fresh score to be approximately Gaussian. The bounded-teacher and score-tail events may be dependent; only the union bound was used.

F.6 All positive times and arbitrary stopping rules

Set $T_F = T(R_F)$, a deterministic finite integer. On $\mathcal{G}_d$, (F.4.4) and (F.5.4) give

$$\inf_{t\ge T_F} \mathrm{CE}_{t,d} \ge \varepsilon_F. \tag{F.6.1}$$

For the finitely many earlier times, apply **Theorem A**. It applies to every fixed finite horizon, every fixed $c > 0$, and all finite positive α, γ. It proves deterministic limits

$$\mathrm{CE}_{t,d} \xrightarrow{P} C_\beta(A_t, Q_t) > 0 \qquad (1 \le t < T_F). \tag{F.6.2}$$

The positive-definite residual covariance in that theorem is valid also in the present separable regime. No finite-MLE result is imported.

Define

$$\varepsilon_{\text{early}} = \frac{1}{2}\min_{1\le t<T_F} C_\beta(A_t, Q_t) > 0, \qquad \varepsilon_* = \min\{\varepsilon_F, \varepsilon_{\text{early}}\} > 0. \tag{F.6.3}$$

The minimum is over a nonempty finite set because $T_F \ge 2$. A finite union bound in (F.6.2), combined with (F.2.11) and (F.6.1), proves the new theorem:

$$\boxed{\mathbb{P}\left\{\inf_{t\in\{1,2,\dots\}} \mathrm{CE}_{t,d} \ge \varepsilon_*\right\} \longrightarrow 1.} \tag{F.6.4}$$

In particular, for **every** sequence of training-measurable stopping rules $\tau_d \in \{1, 2, \dots\}$, with no upper bound on their size,

$$\mathbb{P}(\mathrm{CE}_{\tau_d,d} \ge \varepsilon_*) \longrightarrow 1. \tag{F.6.5}$$

The event in (F.6.4) is common to all rules; the order of its quantifiers is stronger than proving (F.6.5) one rule at a time. Extra randomization of the rule does not affect this pathwise implication. Time zero remains the calibrated constant predictor. A finite unregularized MLE does not exist in this separable regime, so no such object is claimed here.

F.7 Population-risk divergence and a finite optimal-stopping window

The same feature event yields a useful stronger conclusion about risk. Conditional on X, logistic cross-entropy is at least

$$\min\{p(Z), 1-p(Z)\}|S_b|.$$

Use the joint event $\{|S_b| \ge \kappa\|b\|, |Z| \le B_F\}$, whose probability is at least $p_F/2$. Uniformly for all coefficients,

$$\mathcal{R}_F(b) := \mathbb{E}[\ell(Y, S_b) \mid F, b] \ge a_F\|b\|, \qquad \mathcal{R}_F(b) \le \log 2 + L_F\|b\|. \tag{F.7.1}$$

In the lower bound, $\ell(Y,S) = \log(1+e^S) - YS$; this is the same loss as $\ell((2Y-1)S)$ in Section F.1. The upper bound follows from $\ell(Y,S) \le \log 2 + |S|$ and Gaussian Poincare. If b is trained on data, both expectations are fresh-test population quantities conditioned on that training environment.

Combining (F.7.1) and (F.4.2), on one event of probability tending to one,

$$\mathcal{R}_{t,d} \ge \frac{a_F}{D}\left[-\log(e^{\mathfrak{B}_t} - 1)\right]_+ \quad \text{for every } t \ge 1. \tag{F.7.2}$$

This proves uniform late-time divergence: for each fixed $M > 0$, a deterministic finite time exists after which $\mathcal{R}_{t,d} > M$ simultaneously for all later times with probability tending to one. For every $\tau_d \to \infty$ in probability, $\mathcal{R}_{\tau_d,d} \to \infty$ in probability. Empirical risk nevertheless tends to zero by (F.3.5).

In particular, let

$$R_{\text{risk}} = (\log 2 + 1)/a_F, \qquad T_{\text{risk}} = T(R_{\text{risk}}). \tag{F.7.3}$$

Then $\mathcal{R}_{t,d} > \log 2 + 1$ for all $t \ge T_{\text{risk}}$ on $\mathcal{G}_d$, whereas $\mathcal{R}_{0,d} = \log 2$. Therefore the population-risk optimum over the whole GD path is attained before this fixed time:

$$\inf_{t\ge 0} \mathcal{R}_{t,d} = \min_{0\le t<T_{\text{risk}}} \mathcal{R}_{t,d} \quad \text{with probability tending to one.} \tag{F.7.4}$$

Finite-time risk convergence from the main theorem now gives the deterministic characterization

$$\inf_{t\ge 0} \mathcal{R}_{t,d} \xrightarrow{P} \min_{0\le t<T_{\text{risk}}} \mathcal{R}_\beta(A_t, Q_t), \tag{F.7.5}$$

where the time-zero value is $\log 2$. Thus the fixed-time recursion suffices to compute optimal population-risk stopping in this regime: no unproved exchange of infinite time and dimension is used. If the finite limiting minimum has a unique index, a population-risk-minimizing index equals it with probability tending to one. Without uniqueness, the limiting minimizing set remains the correct conclusion.

F.8 Nontrivial learning and the calibration obstruction can coexist

For $\beta > 0$, retain the exact first-step constants

$$\mu_\beta = \mathbb{E}[G(\sigma(\beta G) - 1/2)] > 0, \quad k = a^2\mu_\beta,$$

$$\tau^2 = \frac{\mu_\beta^2 a^2 v}{\gamma} + \frac{a^4 + v^2/\gamma}{4\alpha}, \qquad V = k^2 + \tau^2. \tag{F.8.1}$$

The limiting first-step score is cU with $U = kZ + \tau G$. Its risk $R(c)$ has $R'(0) = -a^2\mu_\beta^2$ and $R''(c) \le V/4$. Hence

$$R(c) \le \log 2 - ca^2\mu_\beta^2 + c^2 V/8.$$

Choosing a fixed

$$0 < c < \min\left\{2\alpha/k_1,\ 4a^2\mu_\beta^2/V\right\} \tag{F.8.2}$$

simultaneously satisfies stability and gives

$$R(c) < \log 2 - \tfrac{1}{2}ca^2\mu_\beta^2. \tag{F.8.3}$$

Thus a positive-step predictor improves population risk by a fixed amount, yet every positive stopping time still has the uniform calibration floor (F.6.4). In particular, a rule allowed to output time zero that achieves a fixed positive risk improvement with probability tending to one must output a positive time with probability tending to one, and therefore cannot attain asymptotic exact calibration. The result is not merely a pathology of steps that never learn.

There is also an explicit common regime with **strict signed overconfidence at the actual risk-optimal first step**. Retain (F.1.3)'s width condition and choose

$$0 < \beta < \frac{v}{a\sqrt{\gamma k_1}}. \tag{F.8.4}$$

Let c_{risk} be the unique minimizer of the first-step Gaussian risk. The scalar theorem in Appendix E proves both strict risk improvement and strictly positive average confidence minus accuracy at this scale, and gives

$$0 < c_{\mathrm{risk}} < \frac{\beta k}{V} \le \frac{a^2\alpha\gamma\beta^2}{v^2} < \frac{\alpha}{k_1} < \frac{2\alpha}{k_1}. \tag{F.8.5}$$

Here $\mu_\beta \le \beta/4$ and $V \ge v^2/(4\alpha\gamma)$. Thus choosing the fixed step $c = c_{\mathrm{risk}}$ satisfies the same stability assumption as the entire-trajectory theorem. For this single strongly overparameterized model, the first iterate strictly improves risk and is overconfident on average, while no positive stopping time attains exact asymptotic calibration. As in the scalar theorem, this is an average signed statement, not overconfidence at every score. Condition (F.8.4) restricts this compatibility corollary only; the calibration floor and population-risk divergence allow every fixed finite β.

F.9 Abstract extension to any uniformly positive separability margin

The row-Gram condition is a concrete sufficient certificate, not a logical necessity for the late-time argument. Suppose instead that, with probability tending to one:

1. The fresh-input feature conditions (F.2.1) hold with common constants.
2. $\|H\|_{\mathrm{op}}/\sqrt{n} \le D_0$.
3. A unit vector u_d has $\min_i s_i h_i^\top u_d \ge \kappa_{\mathrm{sep}} > 0$, with a deterministic constant κ_{sep}.
4. The GD steps satisfy $0 < \eta_- \le \eta_d \le 4/D_0^2$.

The comparator $b_R = (R/\kappa_{\mathrm{sep}})u_d$ and (F.3.4) give

$$\mathcal{L}_n(b^t) \le \frac{1}{t+1} + \frac{\log^2(t+1)}{2\eta_-\kappa_{\mathrm{sep}}^2 t}. \tag{F.9.1}$$

Equation (F.4.1) holds with D_0 in place of D. The remainder of Sections F.4–7 therefore applies verbatim. If the same fixed-time positive calibration theorem is available, all positive stopping times again have a common calibration floor and the population-risk optimum is in a deterministic finite window.

Consequently a separately proved exact-model max-margin limit $\kappa_d \to \kappa_* > 0$ would extend this module to every parameter point covered by that positive-margin theorem. This paragraph is an implication, not an assertion that a particular external universality theorem has already verified the premise for every aspect ratio. At a zero-margin phase boundary, the deterministic crossing time may diverge and this proof does not close the stopping-time quantifier.

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Public repository and source availability

The complete manuscript, six detailed proof appendices, and editable sources are archived in the public Math Brain repository [9]. The paper-specific archive and PDF are given in [10], together with the release's contribution history and verification record.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Code and data availability

The code and data for this paper are at https://github.com/yspennstate/calibration-along-gradient-descent. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [9].

Declaration of generative AI and AI-assisted technologies

The mathematical arguments, literature comparisons, verification, and writing were produced by AI systems under the author's direction. This manuscript consolidates the finite-time proof package of 24 October 2026 and its strongly overparameterized continuation. The author selected the problem and commissioned and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission or publication rests with the author. Internal AI-assisted mathematical reviews and normalization checks are not external peer review or proof-assistant formalization. The accompanying public source archive records the contribution history and checksums of the released files.